Why chose?

A configurable Transformer Workbench for Multi-Top Quark Classification at the LHC

by

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Amsterdam, May 2026

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1

Introduction

For as long as humans have studied nature, they have asked whether matter can be understood in terms of smaller constituents. Ancient Greek thinkers Leucippus and Democritus proposed that all matter is composed of indivisible *atomoi*; a parallel tradition in Indian natural philosophy, the Vaiśeṣika school, developed the concept of *aṇu* — the smallest irreducible unit of substance. During the Islamic Golden Age, scholars such as Ibn Sīnā preserved and extended this Hellenistic heritage, transmitting it into the European Renaissance. Dalton's atomic theory gave the concept quantitative substance, Mendeleev's periodic table revealed underlying order in the elements, and Rutherford's gold-foil experiment demonstrated that atoms contain dense nuclei — a discovery that initiated nuclear physics. The subsequent identification of protons, neutrons, and ultimately quarks culminated in the Standard Model of particle physics: a quantum field theory that describes the fundamental constituents of matter and three of the four known forces with extraordinary predictive precision.

Modern collider experiments extend this tradition of probing matter through scattering, but at energy scales far beyond Rutherford's original experiment. At the Large Hadron Collider (LHC) at CERN, protons are accelerated to near light-speed and brought into collision; the resulting interactions produce short-lived particles that decay almost immediately. No detector records these particles directly. Instead, physicists *reconstruct* them from the signatures they leave in the detector: charged-particle tracks in inner tracking systems, energy deposits in electromagnetic and hadronic calorimeters, displaced decay vertices, and missing transverse energy ($E_{\mathrm{T}}^{\mathrm{miss}}$) inferred from momentum imbalance. Identifying and classifying collision events is therefore an inference problem, not a direct-observation problem, and the complexity of the detector response shapes every step of the analysis.

The Standard Model is, by any measure, a triumph of twentieth-century physics, yet it is not considered the final theory. Compelling open questions remain: the nature of dark matter, the origin of neutrino masses, the observed matter–antimatter asymmetry of the universe, the absence of gravity from the quantum field-theoretic framework, and the so-called hierarchy problem concerning the stability of the Higgs mass under radiative corrections. Precision measurements and searches for rare processes at the LHC provide indirect sensitivity to physics beyond the Standard Model (BSM), even without direct production of new particles. The High-Luminosity LHC (HL-LHC) programme will deliver substantially larger integrated luminosities over the coming decades, granting access to processes with smaller cross-sections than those currently measurable. The frontier is therefore not only at higher energies, but also in the rarest corners of already accessible data.

Among the fundamental particles of the Standard Model, the top quark occupies a special position. It is the heaviest known elementary particle, with a mass close to the electroweak symmetry-breaking scale, and its large Yukawa coupling to the Higgs boson makes top-associated processes particularly sensitive to the Higgs sector and to BSM effects that couple preferentially to mass. This thesis focuses on a class of rare, top-rich processes: four-top-quark production ($t\bar{t}t\bar{t}$) and its dominant backgrounds — $t\bar{t}H$, $t\bar{t}W$, $t\bar{t}Z$, and $t\bar{t}WW$. These events produce complex final states: many hadronic jets, multiple b-tagged jets from top and Higgs decays, possible charged leptons, and substantial $E_{\mathrm{T}}^{\mathrm{miss}}$. Combinatorial ambiguity is severe, since the assignment of reconstructed objects to parent particles is not unique, and the signal cross-section is small relative to the irreducible backgrounds. Accurate event-level classification is therefore central to any analysis targeting these processes.

The core task addressed in this thesis is supervised event-level classification: assigning a scalar discriminant to each collision event such that signal-like and background-like events are separated as reliably as possible. From a physics perspective, the challenge arises from the rarity of the signal, the complexity of the final state, and the many-object combinatorics. From a machine-learning perspective, the challenge is structural: a collision event is a *variable-length, unordered set* of reconstructed objects, each described by a mixture of continuous kinematic features and discrete categorical labels (particle type, charge, b-tag decision). Standard fixed-size or sequence-based architectures impose an artificial structure on this set-valued input, whereas the physical content is permutation-invariant.

Machine learning has become indispensable in high-energy physics for tasks ranging from jet tagging and particle identification to full event reconstruction and classification. The progression from boosted decision trees and shallow neural networks to deep architectures and, more recently, to attention-based models reflects both the growing complexity of analysis tasks and the increasing availability of large simulated datasets. Transformers are attractive for event classification because self-attention operates over all pairs of input objects without imposing an ordering, making the mechanism structurally compatible with the set-like nature of collision events. Pairwise attention weights can, in principle, encode physically meaningful relations: angular separations (ΔR), invariant masses of object pairs, particle-type compatibility, and global quantities such as $E_{\mathrm{T}}^{\mathrm{miss}}$. This structural alignment between the attention mechanism and the physics of multi-object final states motivates a careful, systematic investigation of which specific design choices — input representation, attention biases, positional encoding, tokenisation strategy — actually improve classification performance.

This thesis investigates transformer-based classifiers for rare multi-top quark processes at the LHC. Rather than proposing a single optimised architecture, it develops a configurable workbench in which architectural choices are exposed as orthogonal axes and evaluated through controlled ablations. The central aim is to determine which physics-motivated and machine-learning-motivated design choices improve event classification, as measured primarily by AUROC and complementary efficiency metrics.

1.1. The Standard Model and Top Quark Physics

This section provides the physics theory context for the thesis. It covers the Standard Model particle content and forces, the special role of the top quark, the four-top signal process and its backgrounds, and the sensitivity of rare multi-top processes to BSM physics.

1.1.1. The Standard Model of particle physics

The Standard Model (SM) is a quantum field theory describing the elementary particles and three of the four fundamental forces: the electromagnetic, weak, and strong interactions. Matter consists of twelve fermions — six quarks and six leptons — organised into three generations of increasing mass. The forces are mediated by gauge bosons: the photon (γ) for electromagnetism, the $W^{\pm}$ and Z bosons for the weak interaction, and eight gluons for the strong interaction. The Higgs boson, the quantum of the scalar field that spontaneously breaks electroweak symmetry and endows the W and Z bosons and the fermions with mass, completes the particle content of the theory.

[TODO: Figure: Standard Model particle table]

1.1.2. The top quark

The top quark is the heaviest known elementary particle, with a mass of approximately $172.5\,\mathrm{GeV}$, close to the electroweak symmetry-breaking scale set by the Higgs vacuum expectation value of $246\,\mathrm{GeV}$. Its large Yukawa coupling to the Higgs boson — close to unity — makes the top quark a unique probe of the Higgs sector and of mass-generation mechanisms. Crucially, the top quark decays before it hadronizes, with a lifetime of approximately $5 \times 10^{-25}\,\mathrm{s}$, so its spin and other quantum numbers are transmitted directly to its decay products. Top quarks decay almost exclusively via $t \to Wb$, producing a W boson and a b-quark in essentially every event.

1.1.3. Multi-top production at the LHC

Four-top-quark production ($t\bar{t}t\bar{t}$, hereafter *four-top*) is among the rarest Standard Model processes accessible at the LHC, with a predicted cross-section of approximately $12\,\mathrm{fb}$ at $13\,\mathrm{TeV}$ [1]. The dominant backgrounds to four-top are associated production processes: $t\bar{t}H$, $t\bar{t}W$, $t\bar{t}Z$, and $t\bar{t}WW$, which share similar final-state topologies. Each four-top event produces four b-quark jets from the top decays, four additional jets from hadronic W decays, possible charged leptons, and substantial $E_{\mathrm{T}}^{\mathrm{miss}}$ from neutrinos in leptonic W decays. The resulting jet multiplicities can exceed ten reconstructed objects per event, and the combinatorial ambiguity in assigning jets to parent particles is a central analysis challenge.

[TODO: Figure: Feynman diagrams — QCD and Higgs-mediated four-top production at tree level]

1.1.4. Sensitivity to physics beyond the Standard Model

The four-top process is particularly sensitive to BSM physics because its production rate depends on the top–Higgs Yukawa coupling at the one-loop level and receives contributions from dimension-six effective field theory (EFT) operators involving the top quark. New physics that modifies the top–Higgs coupling or introduces new four-fermion contact interactions can enhance the four-top cross-section by factors of a few without being excluded by other measurements. Precision measurement of the four-top rate thus constrains the Wilson coefficients of top-sector EFT operators independently of direct new-particle searches. Rare multi-top processes therefore occupy a privileged position in the BSM landscape: they are both measurable with HL-LHC luminosity and theoretically clean probes of the top–Higgs sector.

1.2. The ATLAS Experiment and Event Reconstruction

This section describes the experimental apparatus and the data-processing chain that converts raw detector signals into the physics objects used as classifier inputs. It covers the LHC collider, the ATLAS detector subsystems, the object reconstruction pipeline, and the classical analysis strategies that motivate the machine-learning approach.

1.2.1. The Large Hadron Collider

The Large Hadron Collider at CERN is a circular proton–proton (pp) collider with a circumference of $27\,\mathrm{km}$, operating at centre-of-mass energies of $13\,\mathrm{TeV}$ (Run 2) and $13.6\,\mathrm{TeV}$ (Run 3). Proton beams are organised into bunches crossing at four interaction points every $25\,\mathrm{ns}$, delivering peak instantaneous luminosities of order $10^{34}\,\mathrm{cm}^{-2}\mathrm{s}^{-1}$. Each bunch crossing produces multiple inelastic pp interactions simultaneously, a phenomenon known as pile-up, which must be disentangled from the hard-scatter interaction of interest. The HL-LHC upgrade, planned for the late 2020s, will increase instantaneous luminosity by approximately a factor of five, substantially increasing both the dataset size and the pile-up challenge.

[TODO: Figure: LHC schematic]

1.2.2. The ATLAS detector

ATLAS (A Toroidal LHC ApparatuS) is a general-purpose particle detector surrounding one of the four LHC interaction points, designed to measure the full range of SM and BSM processes accessible at LHC energies. The innermost subsystem is the Inner Detector (ID), consisting of a silicon pixel detector, a silicon strip tracker, and a transition radiation tracker, which reconstructs charged-particle trajectories in a $2\,\mathrm{T}$ solenoidal magnetic field. Surrounding the ID are the electromagnetic and hadronic calorimeters, which absorb and measure the energy of photons, electrons, and hadrons; the hadronic calorimeter uses iron absorbers and scintillating tiles in the central region. The outermost system is the Muon Spectrometer, which identifies and measures muons using large toroidal magnets, providing coverage over $|\eta| < 2.7$.

[TODO: Figure: ATLAS detector cutaway]

1.2.3. From detector signals to physics objects

Reconstructing a physics event from raw detector signals proceeds through a sequence of steps, each translating lower-level information into higher-level objects. Charged-particle tracks are reconstructed in the ID from hit patterns in the silicon layers and extrapolated to identify primary and secondary vertices; displaced secondary vertices from b-hadron decays form the basis of b-tagging algorithms. Calorimeter energy deposits are clustered into topological clusters and calibrated to form jets using the anti-k_t algorithm; b-jets are identified by a multivariate tagger exploiting track impact parameters and secondary vertex information. Electrons and muons are identified and calibrated using combined tracking and calorimeter information, while $E_{\mathrm{T}}^{\mathrm{miss}}$ is computed from the negative vector sum of all reconstructed and calibrated object momenta in the transverse plane.

[TODO: Figure: four-top collision evolving through the ATLAS detector (supervisor diagram)]

1.2.4. Event-level classification in classical analyses

Classical analyses separate signal from background using cut-based selections on individual object multiplicities and kinematic variables, followed by multivariate discriminants such as boosted decision trees (BDTs). In four-top analyses, high jet and b-jet multiplicity requirements reduce the dominant $t\bar{t}$+jets background by orders of magnitude while preserving a significant fraction of signal events. BDTs trained on a set of hand-crafted global event variables provide additional separation, but their performance degrades as the number of relevant features grows and correlations between variables become complex. Furthermore, the combinatorial assignment of reconstructed jets to the four top quarks is intractable to solve exactly, so classical approaches either ignore it or rely on dedicated reconstruction algorithms that introduce additional assumptions.

1.3. Machine Learning for High-Energy Physics

This section traces the progression of machine learning in particle physics from early shallow methods through deep learning and, finally, to the attention-based models that motivate this thesis. It covers the historical development, the transformer architecture, its adoption in HEP, and the structural reasons why attention-based models are well-suited to multi-object event classification.

1.3.1. A brief history of machine learning in particle physics

Machine learning has been applied in particle physics since at least the 1990s, when artificial neural networks and Fisher discriminants began supplementing cut-based selections in collider analyses. Boosted decision trees became the dominant paradigm in the 2000s, exploiting their robustness to non-linearities and their tractable training, and remain competitive in many current analyses. The deep learning revolution of the 2010s, driven by convolutional neural networks and large-scale GPU computing, spread quickly into HEP: jet tagging emerged as the primary proving ground, with networks operating directly on calorimeter images or particle lists achieving substantial improvements over BDTs. More recently, graph neural networks and attention-based models have extended the reach of deep learning to irregularly structured inputs, enabling end-to-end event reconstruction and multi-object classification.

1.3.2. Transformers and the attention mechanism

The transformer architecture, introduced by Vaswani et al. in "Attention Is All You Need" [**vaswani2017attention**], replaces recurrence and convolution with a scaled dot-product self-attention mechanism that relates each element of a sequence to every other element simultaneously. Multi-head attention computes several attention functions in parallel and concatenates the results, allowing the model to attend jointly to information from different representation subspaces. The architecture is completed by position-wise feed-forward networks, residual connections, and layer normalisation, all of which have become standard components of modern sequence and set models. Transformers achieved state-of-the-art performance across natural language processing benchmarks and subsequently demonstrated strong transfer to vision, audio, and scientific domains.

[TODO: Figure: Transformer architecture diagram]

1.3.3. Transformers in high-energy physics

The adoption of transformers in HEP accelerated with ParticleNet and its graph-based predecessors, followed by Particle Transformer (ParT), which demonstrated that attention over particle lists consistently outperforms graph-convolutional and convolutional baselines on jet tagging benchmarks. PELICAN introduced Lorentz-equivariant message passing, and OmniJet and related models explored large-scale pre-training on diverse jet datasets. These developments collectively established attention-based architectures as the leading approach for particle-level classification tasks, motivating their application to the more complex problem of full-event classification. The convergence on attention-based models across multiple groups and benchmarks suggests that the structural properties of self-attention, rather than incidental training advantages, are responsible for the gains.

1.3.4. Why transformers suit multi-object event classification

Self-attention is permutation-equivariant by construction: the output for each input token depends on its pairwise relations with all other tokens but not on the order in which they are presented. This property aligns precisely with the physics of collision events, where the set of reconstructed objects has no natural ordering and the relevant information is encoded in pairwise and global relations. Attention weights can be biased by physically motivated pairwise features — angular separations (ΔR), invariant masses, and particle-type compatibility — providing a natural inductive bias without hard-coding specific interaction patterns. These structural considerations directly motivate the design axes explored in this thesis, particularly the input representation choices (Chapter 4) and the physics-informed attention bias families (Chapter 6).

1.4. Thesis Statement

This section articulates the research question, the methodological approach, and the organising principle of the experimental programme.

1.4.1. The intersection of top physics and transformer design

Rare multi-top production and attention-based classifiers are unusually well-matched: the physics produces variable-length, unordered sets of objects with rich pairwise structure, and the transformer processes exactly this kind of input through learned pairwise attention. Despite this structural compatibility, it remains unclear which specific design choices are responsible for performance gains observed in related domains, and whether physics-motivated inductive biases improve over purely data-driven attention for this task. This thesis addresses that gap by asking: which transformer design choices matter for event-level classification of rare multi-top processes, and does physics motivation improve or merely complicate the architecture? The answer requires controlled experiments that isolate individual design decisions rather than co-optimising all parameters simultaneously.

1.4.2. Controlled ablation over a single optimised model

The key methodological choice is to study design decisions through controlled ablations over a single, reproducible base model rather than by comparing independently optimised architectures. Each experiment varies exactly one axis while holding all others at their defaults, so any observed change in AUROC is attributable to that single choice. This avoids the confounding that arises when different architectures differ simultaneously in multiple design dimensions, a common limitation of architecture comparison studies in the literature. The scientific question is therefore not which configuration achieves the highest AUROC, but which design choices matter and whether their effects are consistent with the physics motivation claimed for them.

1.4.3. A configurable workbench for architectural exploration

The thesis organises 33 architectural design choices into orthogonal axes, each exposed as a named parameter in a Hydra configuration tree. Any run in the experiment database can be regenerated from a single CLI command by specifying the relevant axis overrides, ensuring full reproducibility. The axes are grouped into ten thematic groups — covering data treatment, tokenisation, positional encoding, pre-encoder modules, attention and encoder block design, feed-forward network realisation, physics-informed attention biases, classifier head, and model capacity — which map directly to the chapter structure of Part II. This workbench is both the primary engineering deliverable and the experimental substrate for all quantitative claims in the thesis.

1.5. Deliverables

This thesis produces three concrete outputs, each serving a distinct purpose within the overall research programme.

1.5.1. Modular codebase

The primary engineering output is a Hydra-driven Python package in which every architecture variant is reproducible from a single configuration file. Any run appearing in this thesis can be regenerated with a single CLI command.

1.5.2. Experiment database

A Weights & Biases project serves as a structured record of all training runs, storing hyperparameters, metrics, and axis settings in a queryable format. This database is the raw evidence underlying every quantitative claim in the thesis.

1.5.3. Thesis document

This document is a curated, interpretable distillation of the experiment database: not an exhaustive log of all runs, but a set of focused scientific arguments each anchored to a specific axis group.

1.6. Reader's Guide

1.6.1. Structure of the thesis

The thesis is organised in three parts: Part I frames the problem and the workbench (Chapters 2–3); Part II addresses one focused scientific question per chapter, each mapped to a specific axis group (Chapters 4–9); Part III synthesises the findings into a recommended configuration and outlook (Chapters 10–11).

1.6.2. How to read the results

Throughout the thesis, AUROC is the primary metric, reported as a seed-averaged mean with error bars. Axis groups are referred to by their labels (D, A, B, P, H), which correspond directly to config namespaces in the codebase. Each Part II chapter is framed by one of three evaluation lenses — performance, interpretability, or efficiency — which is stated explicitly at the chapter opening.

2

Dataset and Task Definition

3

The Workbench: Architecture and Axes

3.1. Design philosophy

Rather than committing to a single architecture, the workbench exposes every meaningful design choice as a switchable configuration parameter. This makes fair comparison possible by construction: when one axis changes, everything else is held fixed, so any observed difference in AUROC is attributable to that choice alone. The entire architecture is specified by a Hydra configuration tree, where each axis corresponds to a named key in a structured YAML hierarchy. Any run in the experiment database can be regenerated from a single command-line invocation.

3.2. The base transformer

The base model follows the standard transformer encoder pipeline: particle tokens are embedded into a fixed-dimensional space, passed through a stack of attention-plus-FFN blocks, aggregated into a single event-level vector by a pooling operation, and classified by a head network. This skeleton is fixed across all experiments; the axes control what happens inside each stage.

Each particle token, a raw kinematic vector plus an optional particle-type identifier, is projected into the model dimension d by a linear embedding layer. The encoder consists of L identical blocks, each containing a multi-head self-attention layer followed by a position-wise feed-forward network, with residual connections and normalisation at positions controlled by axes A1 and A2. The pooled vector is passed to a classifier head that produces the final logit.

3.3. Axis groups

The axes are organised into ten groups. Groups G and D define the experimental conditions and data treatment. Groups T, E, P, A, F, B, and C define the architecture. Group H sets model capacity. Three cross-cutting shared parameter blocks (§K, §M, §S) are consumed by multiple modules simultaneously and are documented at the end of this section. Groups §R and §L cover training protocol and logging respectively and are not architectural axes.

G Study framing

G axes define what is being classified and with which model family. They are not architectural choices and are not varied in the ablation programme; they set the experimental conditions under which all other axes are evaluated.

G01 – Task type. Selects the training loop: supervised classification or anomaly detection (autoencoder, GAN-AE, diffusion-AE). This thesis focuses exclusively on supervised classification; the anomaly detection loops are available in the workbench but not studied here.

G02 – Model family. Selects the model architecture family: transformer, MLP, BDT, or autoencoder. All Part II chapters use the transformer family; the BDT appears only in Ch. 8 as a surrogate model for configuration prediction, not as a classifier.

G03 – Classification task. Defines which processes are treated as signal and background. Five processes are available: four-top-quark production ($tttt$), associated production with a Higgs boson (ttH), with a W boson (ttW), with two W bosons ($ttWW$), and with a Z boson (ttZ). Three task definitions are used in this thesis: $tttt$ vs. combined background ($ttH + ttW + ttWW + ttZ$) as the primary task, $tttt$ vs. ttH as a harder secondary task, and all-five multiclass as an exploratory task.

D Data treatment

D axes control what the model sees before any architectural processing.

D01 – Feature set. Selects which kinematic features are included in each token. The default is the four-feature set (E, Pt, η, ϕ); energy can be omitted to test whether the model relies on explicit E or whether Pt, η, and ϕ already provide sufficient kinematic information.

D02 – MET treatment. Controls whether missing transverse energy is excluded or included as an event-level global. Tests how much the invisible neutrino system contributes beyond the particle tokens.

D03 – Token ordering. Sets whether tokens are presented in input order, sorted by decreasing Pt, or randomly shuffled per event. Tests whether the model exploits ordering as an implicit positional signal.

T Tokenizer

T axes control how raw particle features are converted into token embeddings.

T1 – Tokenizer family. Selects between `raw` (kinematics only), `identity` (kinematics plus an explicit particle-type label), and `binned` (discrete token, no continuous features). Tests whether explicit particle identity is necessary or recoverable from kinematics alone.

T1-a – PID embedding mode. When T1 = `identity`, selects how the particle-type label is encoded: `learned`, `one_hot`, or `fixed_random`. Tests whether particle identity needs a learned geometry or only a categorical code.

T1-b – PID embedding dimension. Sets the capacity of the PID embedding to 8, 16, or 32 dimensions. Tests the trade-off between representational capacity and overfitting for the type sub-embedding.

E Positional encoding

E axes control how position information is injected into the token stream.

E1 – PE type. Selects among `none`, `sinusoidal`, `learned`, and `rotary`. Tests whether injecting position information helps a set-based classifier at all.

E1-a – PE space. Controls whether the positional signal is applied before or after the input projection. Prerequisite: E1 $\in$ {sinusoidal, learned}.

E1-a1 – PE dimension mask. Allows the positional encoding to act selectively on a subset of feature dimensions. Prerequisite: E1-a = `token`.

P Pre-encoder modules

P axes add optional modules applied to raw particle features before the main transformer encoder. Both require T1 $\in$ {raw, identity}.

P1 – Nodewise mass patch. Optionally enriches each token with invariant-mass features computed from its k-nearest neighbours in Lorentz space, following Li et al. Tests whether explicit local mass structure before the encoder improves token representations. Requires D01 to include energy (index 0).

P2 – MIA pre-encoder. Optionally inserts MIParT-style interaction-attention blocks before the main encoder, where attention is driven by pairwise interaction features rather than query-key similarity. Tests whether physics-driven message passing as a pre-encoder stage adds information beyond standard self-attention.

P2-a – MIA placement. Controls whether MIA blocks are prepended or appended relative to the main encoder. Prerequisite: P2 = `true`.

A Attention and encoder block

A axes control the internal design of each encoder block.

A1 – Normalisation policy. Selects pre-norm, post-norm, or NormFormer placement of layer normalisation within each block. Tests whether normalisation position affects training stability for particle-level inputs.

A2 – Normalisation type. Chooses between LayerNorm and RMSNorm. Tests whether removing the mean-subtraction step improves efficiency without degrading performance.

A3 – Attention type. Selects standard multi-head attention or differential attention, which computes the difference of two softmax maps to suppress diffuse attention patterns. Tests whether this mechanism transfers from NLP to particle classification.

A3-a – Differential bias mode. When A3 = `differential`, controls how additive physics biases interact with the two attention branches: `none`, `shared`, or `split`. Prerequisite: A3 = `differential`.

A4 – Attention-internal normalisation. Optionally applies LayerNorm or RMSNorm per head after the weighted sum, before head merging. Tests whether per-head normalisation stabilises attention output scales. Applies to both standard and differential attention.

A5 – Causal masking. Optionally applies an autoregressive mask to the attention matrix. Tests whether imposing sequential structure on an inherently unordered set of particles hurts, helps, or is neutral.

F Encoder FFN realization

F axes control the feed-forward network inside each encoder block. The effective realization follows a priority rule: MoE $>$ KAN $>$ standard.

F1 – FFN realization. Selects between a standard two-layer MLP, a Kolmogorov–Arnold Network (KAN), and a Mixture-of-Experts (MoE) FFN. Tests whether learnable spline activations or conditional computation improve per-token transformations.

F1-a – KAN FFN variant. When F1 = `kan`, selects among `hybrid`, `bottleneck`, and `pure` architectures. Prerequisite: F1 = `kan`.

F1-b – MoE scope. Controls whether expert routing applies to encoder FFNs, the classifier head, or both. Prerequisite: F1 = `moe`.

B Physics-informed attention biases

B axes inject physics structure as additive biases on the attention logits. All families require T1 $\in$ {raw, identity} and share a zero-initialised scalar gate, so adding any bias leaves the baseline unchanged at initialisation. Multiple families can be active simultaneously using + notation.

B1 – Bias activation set. The master switch selecting which bias families are active: `lorentz_scalar`, `typepair_kinematic`, `sm_interaction`, `global_conditioned`, or combinations thereof.

B1-L1 – Lorentz feature set. Selects the subset of pairwise Lorentz-invariant features fed into the bias network, from the minimal set {m^2, ΔR} through the MIParT basis to the full feature set. Active when `lorentz_scalar` $\in$ B1.

B1-L2 – Lorentz MLP type. Selects whether the network mapping Lorentz features to bias scores uses a standard MLP or a KAN. Active when `lorentz_scalar` $\in$ B1.

B1-L5 – Lorentz sparse gating. Adds per-feature sigmoid gates providing a direct measure of learned feature importance. Active when `lorentz_scalar` $\in$ B1.

B1-T2 – Type-pair initialisation. Controls how the learnable $N_{\text{type}} \times N_{\text{type}}$ interaction table is initialised: from scratch, from a binary SM mask, or from fixed SM coupling constants. Active when `typepair_kinematic` $\in$ B1.

B1-T3 – Type-pair freeze. Controls whether the type-pair table is frozen after initialisation or remains trainable. Tests whether the table is better used as a fixed physics prior or a learnable parameter. Active when `typepair_kinematic` $\in$ B1.

B1-S1 – SM interaction mode. Selects the fidelity of the SM interaction prior: `binary`, `fixed_coupling`, or `running_coupling`. Active when `sm_interaction` $\in$ B1.

B1-G1 – Global-conditioned mode. Controls how MET modulates pairwise attention biases: as a global scale factor or as a directional signal. Requires D02 = `true`. Active when `global_conditioned` $\in$ B1.

C Classifier head

C axes control the final mapping from the pooled event representation to class logits.

C1 – Head realization. Selects between a `linear` head, a `kan` head, and a `moe` head. Tests whether specialised nonlinearities or conditional computation at the decision boundary improve discrimination.

C2 – Pooling strategy. Selects how the token sequence is aggregated into a single event-level vector: `cls` token, `mean` pooling, or `max` pooling. Placed here because pooling is logically part of the readout stage rather than the encoder block.

H Model-size hyperparameters

H axes set model capacity and are not architectural design choices in the sense of the other groups. They are tracked separately and are the primary axis of variation in Ch. 9. The convenience key `model_size_key` bundles dim (d), depth (L), heads, and FFN hidden dim into named presets spanning roughly 50k to 4M parameters.

Cross-cutting blocks: §K, §M, §S

Three shared parameter blocks are consumed simultaneously by multiple modules. §K (KAN hyperparameters: grid size, spline order, grid range) affects every active KAN module, including the encoder FFN, classifier head, and Lorentz bias network. §M (MoE hyperparameters: number of experts, top-k, routing level) affects all active MoE sites. §S (shared pairwise backbone) computes the pairwise feature matrix F_{ij} once per forward pass and shares it across all active B1 families and the MIA pre-encoder.

3.4. Experimental methodology

Each axis is studied in isolation: one axis is varied across its settings while all others are held at their defaults, forming an orthogonal sweep. This ensures observed differences are attributable to the axis under study and keeps the number of required experiments linear in the number of axes rather than exponential.

Every result reported in this thesis is averaged over at least three independently seeded runs, with error bars representing the standard deviation across seeds. The area under the ROC curve (AUROC) is the primary evaluation metric throughout: it measures the probability that the classifier assigns a higher score to a signal event than to a background event, and is threshold-independent. Signal efficiency at fixed background rejection is reported as a complementary metric in selected chapters.

All runs are tracked in a Weights & Biases project where every axis setting, hyperparameter, and evaluation metric is logged under a structured key scheme mirroring the axis reference directly.

Part II

Thesis Questions

4

Best Input Representation

Before the transformer processes a single attention operation, several choices have already determined the information it receives. These choices concern *how* each particle is represented as a token, and *what* the token sequence contains. This chapter evaluates six such choices — the tokenizer family (T1), the particle-identity (PID) embedding mode (T1-a), the PID embedding dimension (T1-b), the kinematic feature set (D01), whether missing transverse energy enters as an additional token (D02), and the ordering of tokens in the input sequence (D03) — and asks which of them affect classification performance on the 4-tops-versus-background task.

The chapter is structured as follows. Section 4.1 describes the shared experimental setup and evaluation metric. Section 4.2 presents the tokenizer family comparison, which is the primary result of this chapter. Section 4.2.2 examines the sub-choices within the identity tokenizer (T1-a and T1-b). Section 4.3 covers the data-treatment axes (D01, D02, D03). Section 4.4 synthesises the findings and establishes the baseline configuration carried forward into subsequent chapters.

The one-sentence answer is: the tokenizer family dominates all other choices, with the identity tokenizer outperforming kinematics-only alternatives by a large margin; fine-grained choices within that family and all data-treatment axes are null or near-null results at this model scale.

4.1. Experimental Setup

Two experiments are reported here. **Experiment 4A** sweeps the tokenizer family (T1) and, within the identity family, the PID embedding mode (T1-a) and embedding dimension (T1-b). **Experiment 4B** fixes the tokenizer at identity and sweeps the three data-treatment axes (D01, D02, D03) in a full $2 \times 2 \times 2$ factorial design.

All axes not under study are held at canonical defaults throughout: positional encoding (E1) is sinusoidal in model space; attention biases (B1) are disabled; pooling (C2) uses a CLS token; the FFN (F1) is the standard implementation. The model size is the `d64_L4` baseline (H01 = 64, H02 = 4), used uniformly across all runs in this chapter. Every condition is replicated with three seeds (R05 $\in$ {42, 123, 314}).

The primary evaluation metric is the area under the receiver operating characteristic curve (AUROC) computed on the held-out test set. All metric values are drawn from `eval_v2/test_auroc` in the W&B export, which is recorded at training time and is not affected by the subsequent re-inference step. Reported means and standard deviations are over the three seeds unless stated otherwise; with $n = 3$ seeds per condition, these figures are indicative and should not be interpreted as tight confidence intervals.

Table 4.1 and Table 4.2 summarise the sweep designs for the two experiments.

Axis	Values swept	n_{configs}
T1 — tokenizer family	`raw`, `binned`	2
T1 + T1-a — PID mode	`one_hot`, `fixed_random`, `learned`	3
T1 + T1-b — PID dim	8, 16, 32	3
Unique configurations		11
Seeds per configuration		3
Total runs		**33**

Table 4.1: Experiment 4A sweep design. Axes T1-a and T1-b apply only to the `identity` tokenizer family. All other axes are held at canonical defaults: D01 = `[0,1,2,3]`, D02 = `false`, D03 = `input_order`.

Axis	Values swept	n_{configs}
D01 — feature set	`[0,1,2,3]`, `[1,2,3]`	2
D02 — MET treatment	`false`, `true`	2
D03 — token ordering	`input_order`, `shuffled`	2
Unique configurations (2^3)		8
Seeds per configuration		3
Total runs		**24**

Table 4.2: Experiment 4B sweep design ($2 \times 2 \times 2$ full factorial). Fixed: T1 = `identity`, T1-a = `learned`, T1-b = 8. These tokenizer settings are confirmed from the W&B export; all 24 rows use this configuration.

4.2. Tokenizer Family

The choice of tokenizer family (T1) is the most consequential design decision examined in this chapter. Three families are compared. The `raw` tokenizer maps each particle's continuous kinematics (E, p_T, η, ϕ) directly to the model dimension via a shared linear projection; no particle-type information is provided. The `identity` tokenizer augments the kinematic projection with an explicit PID embedding, which encodes the discrete particle type label (jet, b-jet, electron, muon, photon, etc.) and concatenates it with the projected kinematics. The `binned` tokenizer discards continuous features entirely, representing each particle solely by its histogram-binned kinematic cell index as a discrete token.

Figure 4.1 shows the mean test AUROC for each family. The `identity` tokenizer achieves a mean AUROC of approximately 0.842 (evaluated on cells with embedding dimension ≥ 16, where the learned-mode confound described in Section 4.2.2 is absent; $n = 21$ seeds), compared with 0.804 ± 0.001 for `raw` ($n = 3$) and 0.725 ± 0.002 for `binned` ($n = 3$). The gap between `identity` and `raw` is $+0.038$ AUROC; the gap between `identity` and `binned` is $+0.117$.

Figure 4.2 shows the ROC curves for each family. The identity advantage is present uniformly across the full operating range: at every background rejection threshold, the identity tokenizer achieves higher signal efficiency than the raw tokenizer, and both substantially outperform the binned tokenizer.

The training dynamics in Figure 4.3 add interpretive context. The `binned` tokenizer reaches a validation AUROC peak near epoch 4 and subsequently collapses, behaviour consistent with rapid memorisation of the discrete token vocabulary followed by overfitting once no additional generalisation signal remains. The `raw` tokenizer converges stably but plateaus at a lower level. The `identity` tokenizer converges to a higher plateau with a gradual, mild decline beyond the optimum, consistent with moderate overtraining over 50 epochs.

Interpretation. The result is consistent with a physics-motivated expectation: particle type carries information that is not fully recoverable from the four-vector alone, particularly for particles with overlapping kinematic distributions (e.g. light jets and b-jets at similar p_T and η). By providing an explicit PID label, the identity tokenizer allows the model to apply type-specific attention patterns and feature combinations that the raw tokenizer cannot form. The binned tokenizer is the weakest family because it discards not only the type label but also the continuous kinematic resolution; only coarse bin membership remains.

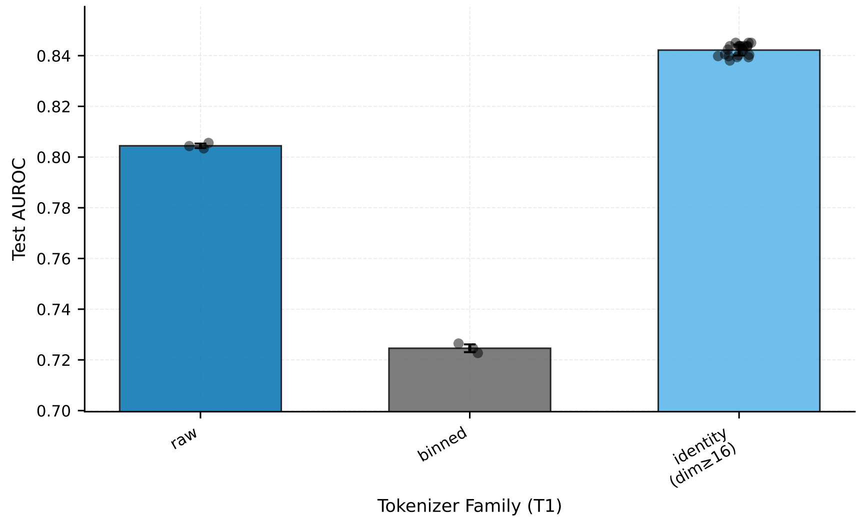

Figure 4.1: Mean test AUROC by tokenizer family (T1). Error bars show one standard deviation over seeds. The `identity` bar pools cells with embedding dimension ≥ 16 ($n = 21$ seeds) to exclude the `learned`/dim=8 confound discussed in Section 4.2.2. `raw` and `binned` each have $n = 3$.

4.2.1. Failure Analysis: Where PID Changes the Outcome

To quantify where the tokenizer gap originates, each event in the test set ($N = 30{,}207$) is classified by both the best `raw` model (AUROC = 0.8056) and the best `identity` model (`one_hot`, AUROC = 0.8451), using a threshold of 0.5 on the signal score.[1]

Figure 4.4 shows the per-event outcome breakdown. Of the $30{,}207$ test events, $20{,}289$ (67.2%) are classified correctly by both tokenizers, and $5{,}396$ (17.9%) are wrong for both, defining an irreducible difficulty floor that reflects events where neither tokenizer has sufficient information to classify correctly. The remaining events reveal the tokenizer effect: $2{,}713$ events (9.0%) are correctly classified by the identity tokenizer but not by the raw tokenizer (PID helped), while $1{,}809$ events (6.0%) are correctly classified by the raw tokenizer but not by the identity tokenizer (PID hurt). The net gain from the explicit PID label is $+904$ events, or $+3.0\%$ of the test set.

The 6.0% of events where the raw tokenizer outperforms the identity tokenizer warrants a brief remark. These events are not evidence that PID is harmful in general; rather, they reflect the stochastic nature of threshold classification and the fact that no tokenizer achieves perfect classification on all events. For some events, the kinematic features alone may provide a cleaner signal than the combination of kinematics and a noisy or ambiguous type label.

4.2.2. PID Embedding Mode and Dimension

Within the identity tokenizer family, two further choices determine how the PID label enters the model: the embedding mode (T1-a) and the embedding dimension (T1-b). This subsection examines these sub-choices, which have a substantially smaller effect than the tokenizer family itself.

Figure 4.5 shows the mean test AUROC by PID embedding mode, pooled over all embedding dimensions. The marginal means are: `one_hot` $= 0.8442 \pm 0.0006$ ($n = 3$), `fixed_random` $= 0.8410 \pm 0.0021$ ($n = 9$), `learned` $= 0.8420 \pm 0.0017$ ($n = 27$). The apparent underperformance of `learned` relative to `one_hot` in this marginal view is a confound: the `learned` sample is dominated by dimension-8 runs, which perform anomalously.

The heatmap in Figure 4.6 resolves this confound by showing AUROC as a joint function of mode and dimension. The per-cell results from Experiment 4A (D01 = `[0,1,2,3]`, D02 = `false`, D03 = `input_order`) are:

- `learned`, dim=8: AUROC $= 0.8431 \pm 0.0007$ ($n = 3$)

[1] W&B run identifiers: raw — `run_20260410-160141_ch4_input_repr_exp4a_1_raw_job002`; identity — `run_20260410-162037_ch4_input_repr_exp4a_2_job010`.

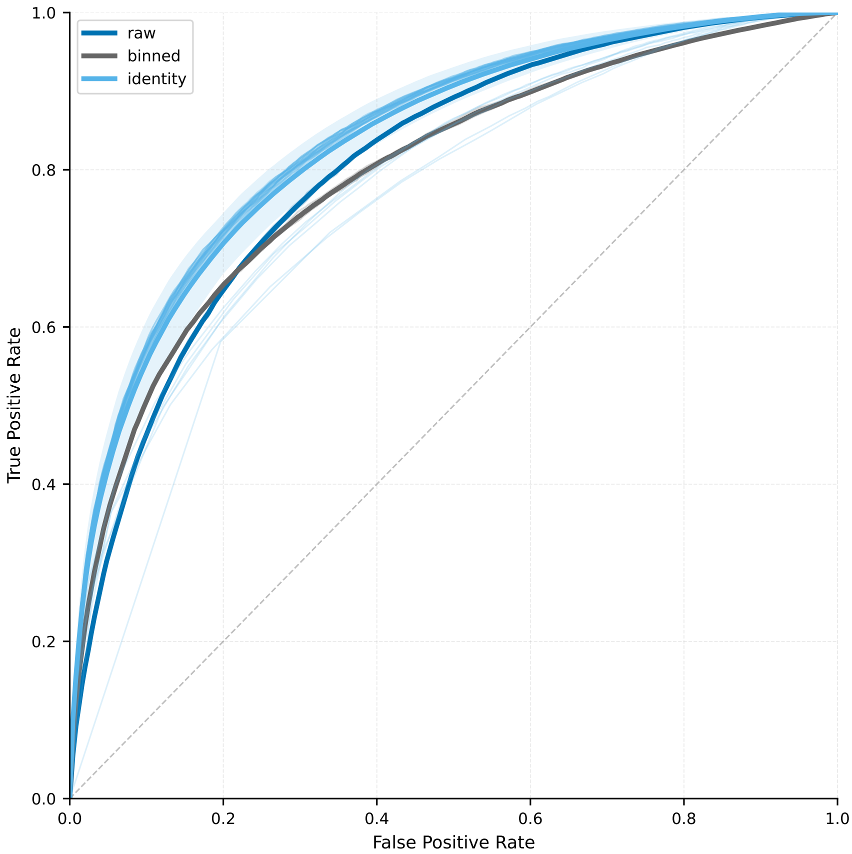

Figure 4.2: ROC curves by tokenizer family (T1). Individual seed curves are shown for `raw` and `binned`; the mean curve is shown for `identity`. The performance ordering is consistent across all background rejection values.

- `learned`, dim=16: AUROC $= 0.8428 \pm 0.0006$ ($n = 3$)
- `learned`, dim=32: AUROC $= 0.8402 \pm 0.0004$ ($n = 3$)
- `one_hot`, num_types: AUROC $= 0.8442 \pm 0.0006$ ($n = 3$)
- `fixed_random`, dim=8: AUROC $= 0.8435 \pm 0.0009$ ($n = 3$)
- `fixed_random`, dim=16: AUROC $= 0.8400 \pm 0.0020$ ($n = 3$)
- `fixed_random`, dim=32: AUROC $= 0.8397 \pm 0.0004$ ($n = 3$)

The spread across all cells is ~ 0.004 AUROC, which is within two standard deviations of the seed spread for individual cells. No mode-dimension combination is strongly superior to any other at $n = 3$ seeds.

Interpretation. The result suggests that the model is not strongly sensitive to whether the PID label is encoded as a trainable embedding, a fixed random projection, or a discrete one-hot vector. All three modes provide a representation that is sufficiently distinct per particle type for the transformer to exploit. The `one_hot` mode at dimension equal to the number of particle types (num_types) is the most parameter-efficient choice, as it requires no additional trainable weights and is insensitive to the embedding dimension hyperparameter. This mode is adopted as the baseline for subsequent chapters, paired with the default dimension setting.

4.3. Data Treatment

Having established the tokenizer family and PID encoding, this section examines three choices about the content and arrangement of the token sequence: whether energy is included in the feature vector (D01), whether MET enters as an additional token (D02), and whether tokens are presented in

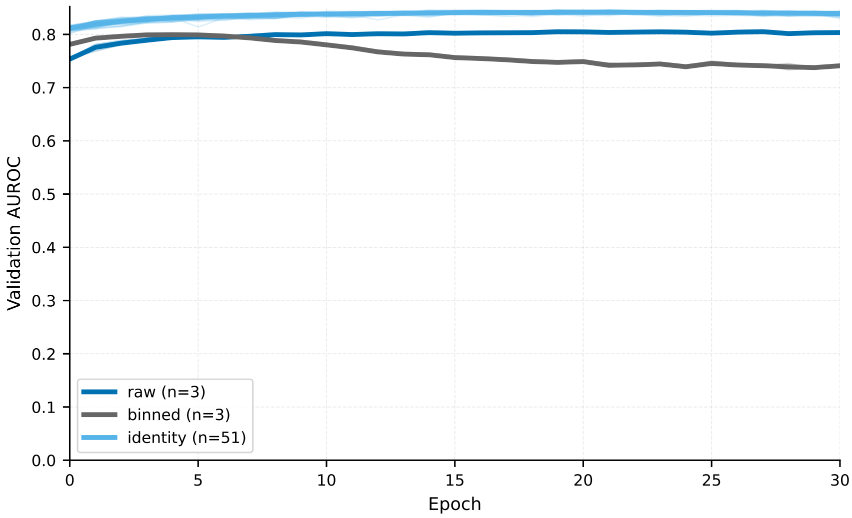

Figure 4.3: Validation AUROC over training epochs by tokenizer family (T1). The `binned` tokenizer peaks near epoch 4 and collapses, indicating rapid overfitting. `raw` converges stably but at a lower plateau. `identity` converges to the highest plateau and degrades only slowly beyond the optimum. The x-axis is cropped to 30 epochs to focus on the convergence region.

their original reconstruction order or shuffled (D03). All results in this section are from Experiment 4B (Table 4.2), which uses the identity tokenizer with T1-a = `learned` and T1-b = 8 throughout.

4.3.1. Energy Feature (D01)

The feature-set axis (D01) compares two configurations: the full four-vector (E, p_T, η, ϕ) with index set `[0,1,2,3]`, and the three-vector (p_T, η, ϕ) with index set `[1,2,3]`, which omits the energy component.

Figure 4.7 shows the result. Removing energy yields AUROC $= 0.8424 \pm 0.0010$ versus 0.8431 ± 0.0007 with energy, a difference of -0.0007. This difference is smaller than the within-condition standard deviation and cannot be distinguished from seed noise at $n = 3$.

The null result is physically plausible: for massless or near-massless particles, energy is approximately $E \approx p_T \cosh \eta$, so p_T and η together recover much of the energy information. The genuinely mass-sensitive component — relevant for reconstructing invariant masses of top decay sub-systems — may not be detectable at this model capacity and sample size. The full four-vector (`[0,1,2,3]`) is retained in the chapter baseline because it carries no performance cost and preserves the option to use the nodewise mass module (P1) in subsequent chapters, which requires the energy feature.

4.3.2. MET Treatment (D02)

Missing transverse energy (MET) represents the net transverse momentum carried by particles that are not detected, primarily neutrinos from W-boson decays. In 4-tops events, which contain at least four top quarks with multiple semi-leptonic or fully hadronic decay chains, MET provides information about the neutrino system. The MET treatment axis (D02) tests whether including MET as an additional token improves classification.

Figure 4.8 shows the marginal comparison across D01 and D03. With MET included (D02 = `true`), the mean AUROC is 0.8423 ± 0.0019 ($n = 12$); without MET, it is 0.8415 ± 0.0017 ($n = 12$). The difference is $+0.0008$ in favour of including MET, which is well within the pooled standard deviation and is not statistically distinguishable from zero at this sample size. Restricting to the `input_order` condition only (D03 = `input_order`, $n = 6$ per MET condition), the delta is $+0.0007$, consistent with the full marginal result.

This is a null result: neither benefit nor harm from MET inclusion is measurable in Experiment 4B. The result is inconsistent with an earlier, superseded figure of -0.026 that appeared in a prior version of the analysis; that figure has been traced to a data-loading bug in the re-inference pipeline, in which the

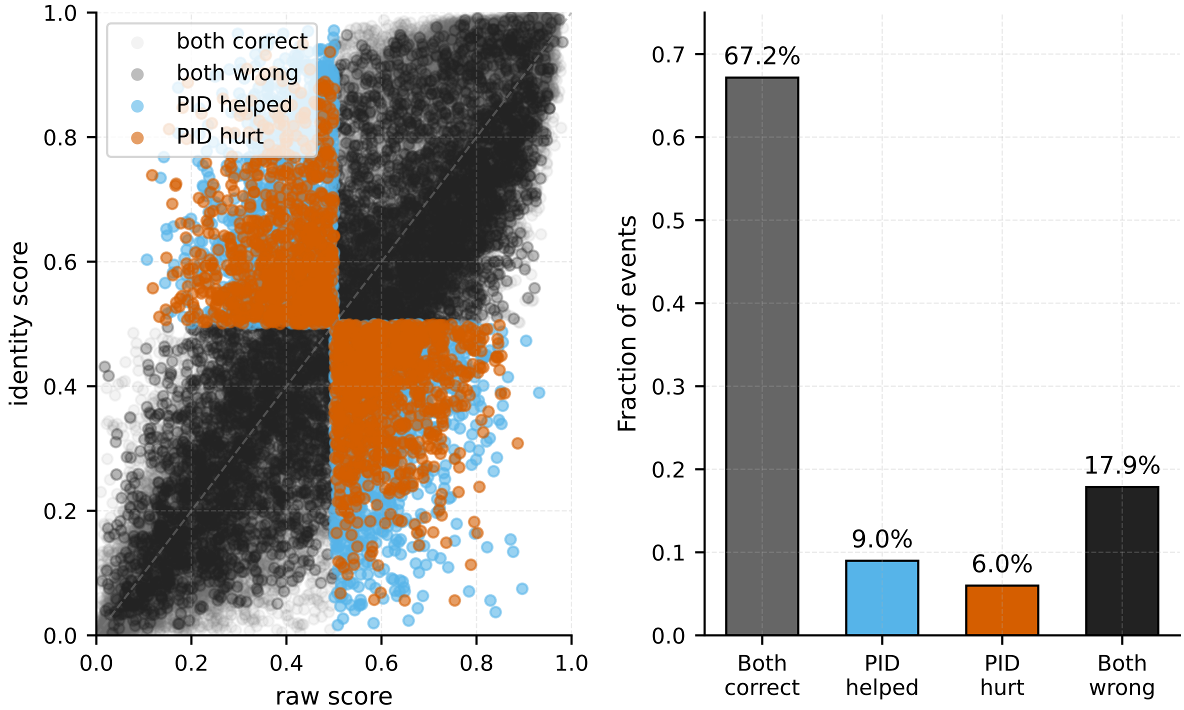

Figure 4.4: Per-event outcome comparison between the best `raw` and best `identity` tokenizer models on the test set ($N = 30{,}207$). *Left:* scatter of per-event signal scores; colours indicate outcome category. Events above the diagonal (identity score $>$ raw score) tend to benefit from the PID label. *Right:* fraction of events in each category. The net asymmetry (9.0% PID-helped vs. 6.0% PID-hurt) accounts for the identity tokenizer's overall AUROC advantage.

validation dataset was not correctly augmented with the MET token for MET=True models. The correct values are those in Table 4.3, drawn directly from `eval_v2/test_auroc` as logged at training time.

The PE-disruption hypothesis — that MET, appended at a fixed token position, disrupts the sinusoidal positional encoding (E1) by occupying a slot that the model otherwise associates with a specific particle type — is a plausible post-hoc explanation that has not been tested experimentally. Such a test would require a dedicated experiment crossing D02 with E1 = `none` (no positional encoding) to isolate any PE-mediated effect. No such experiment has been conducted. MET is excluded from the chapter baseline (D02 = `false`) primarily because it provides no measured benefit; its re-introduction via physics-informed attention biases (B1-G1 = `met_direction`) is examined in a subsequent chapter.

4.3.3. Token Ordering (D03)

Self-attention is permutation-equivariant by construction: without positional encoding, the output is invariant to any permutation of the input tokens. However, all runs in this chapter use sinusoidal positional encoding (E1 = `sinusoidal`) in model space (E1-a = `model`). The token ordering axis (D03) tests whether the choice between preserving the original reconstruction order (`input_order`) and randomly shuffling the sequence (`shuffled`) affects AUROC.

Figure 4.9 shows the marginal comparison restricted to D02 = `false` conditions, which removes any potential interaction with the MET token. `input_order` yields a mean AUROC of 0.8428 ± 0.0008 ($n = 6$); `shuffled` yields 0.8402 ± 0.0012 ($n = 6$). The gap of -0.0026 (shuffled vs. input_order) is approximately 2–$3\times$ the standard deviation of the individual conditions and is consistent across both D01 values.

Table 4.3 shows the full $2 \times 2 \times 2$ factorial mean AUROC table for all D01, D02, D03 combinations. The ordering effect is present under both MET conditions, suggesting it is not mediated by an interaction between D03 and D02.

Attribution. The ordering effect is consistent with the sinusoidal positional encoding providing useful information when tokens are presented in a consistent order, and that information being destroyed by

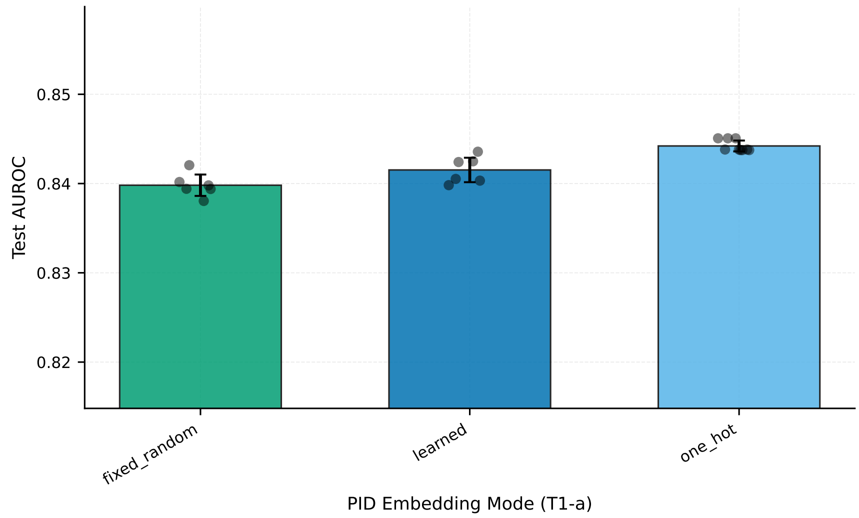

Figure 4.5: Mean test AUROC by PID embedding mode (T1-a), pooled over all T1-b embedding dimensions. The apparent gap for `learned` reflects the dominance of dimension-8 cells in the `learned` sample. Figure 4.6 shows that at dimension ≥ 16, all three modes are within 0.003 AUROC of each other.

D01	D02	D03	mean AUROC	std	n
`[0,1,2,3]`	False	input_order	0.8431	0.0007	3
`[0,1,2,3]`	False	shuffled	0.8398	0.0012	3
`[0,1,2,3]`	True	input_order	0.8434	0.0015	3
`[0,1,2,3]`	True	shuffled	0.8420	0.0008	3
`[1,2,3]`	False	input_order	0.8424	0.0010	3
`[1,2,3]`	False	shuffled	0.8406	0.0013	3
`[1,2,3]`	True	input_order	0.8436	0.0004	3
`[1,2,3]`	True	shuffled	0.8401	0.0021	3

Table 4.3: Full $2 \times 2 \times 2$ factorial AUROC table for Experiment 4B. Each cell reports the mean and standard deviation over three seeds. The token ordering effect (D03) is consistent across all D01 and D02 conditions.

random shuffling. However, this attribution requires a control experiment with E1 = `none` (no positional encoding) to confirm. If the ordering effect disappears under E1 = `none`, the effect is mediated by the positional encoding. If it persists, the transformer's residual stream or the classification head is sensitive to token position through some other mechanism. This control experiment has not been run. The thesis states the ordering effect as a measured result and the PE attribution as a hypothesis.

The reconstruction software produces tokens in an order that is not p_T-sorted but reflects the detector reconstruction sequence, which may carry implicit correlations between position and particle type or kinematics. If such correlations exist, the sinusoidal PE would be implicitly encoding a type-proxy, which is a legitimate, if implicit, inductive bias. Characterising this ordering is a direction for future work.

4.4. Discussion

This section synthesises the results of Experiments 4A and 4B and establishes the baseline configuration for subsequent chapters.

Ranking of effects. The tokenizer family (T1) dominates all other choices by a large margin. The identity-versus-raw gap of $+0.038$ AUROC ($+0.117$ identity versus binned) is an order of magnitude larger than any data-treatment effect. Within the identity family, PID mode (T1-a) and dimension (T1-b) effects are at most ~ 0.004 AUROC, comparable to seed-spread. All three data-treatment axes produce effects below 0.003 AUROC in this experiment; none is distinguishable from noise with certainty at $n = 3$ seeds.

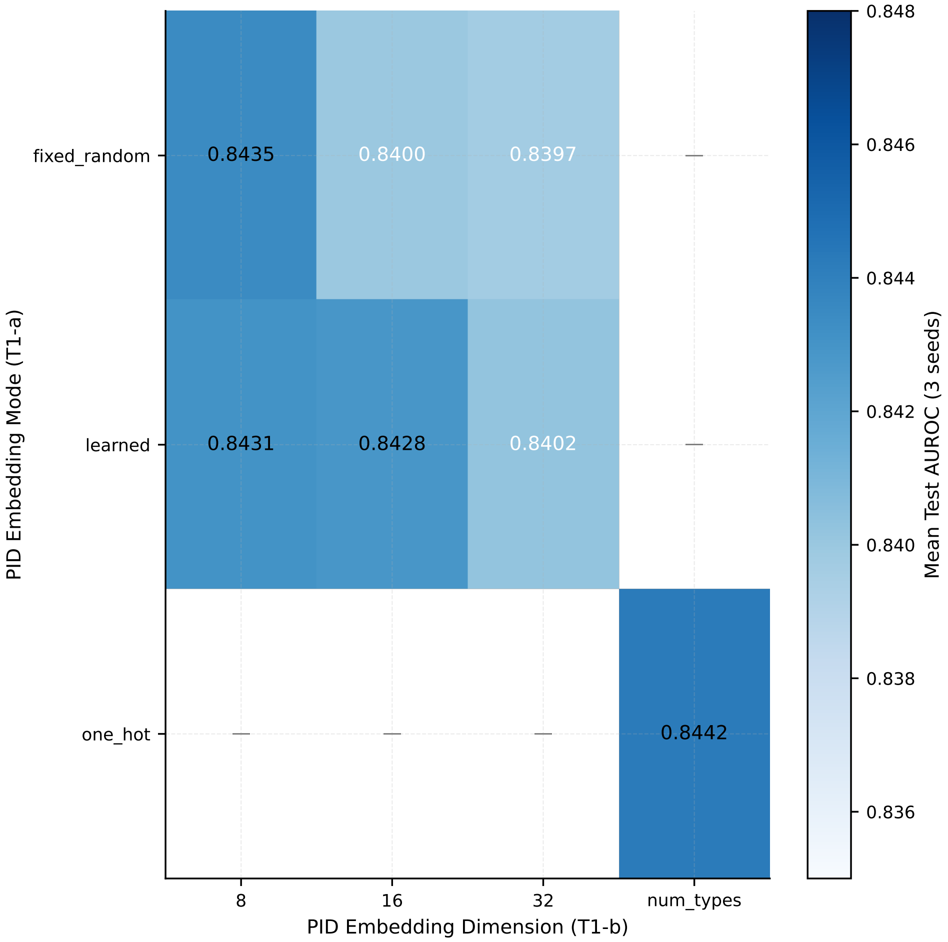

Figure 4.6: Mean test AUROC as a function of PID embedding mode (T1-a) and embedding dimension (T1-b), for the `identity` tokenizer. `one_hot` is invariant to dimension by construction (dimension is set to the number of particle types). All cells show similar performance; the total spread is ~ 0.004 AUROC.

The narrative arc of this chapter is therefore: *choose the identity tokenizer; thereafter, fine-grained choices do not matter at this model scale and seed count.*

Limitations. Three limitations are worth stating explicitly.

First, only three seeds per condition are available. The D03 ordering effect (-0.0026) is the largest of the data-treatment results and is suggestive of a real effect, but would require $n \geq 6$ per condition to approach 95% confidence given typical seed-spread variability of ~ 0.001 AUROC. The D01 and D02 results (< 0.001) are almost certainly null results, but cannot be proven so with the current seed count.

Second, the attribution of the D03 ordering effect to the sinusoidal positional encoding has not been confirmed. An E1 = `none` control is the minimum experiment required. Similarly, the PE-disruption hypothesis for D02 is untested.

Third, Experiment 4B uses T1-a = `learned` and T1-b = 8 as the fixed tokenizer configuration, which is not the best-performing individual cell identified in Experiment 4A. The D02 and D03 effects may differ under `one_hot` tokenization; this cross-check has not been performed.

Chapter baseline. Table 4.4 records the axis values chosen as the baseline for all subsequent chapters. Choices follow the criterion of maximising expected performance while minimising hyperparameter sensitivity.

These six choices define the canonical input representation for the transformer workbench as used

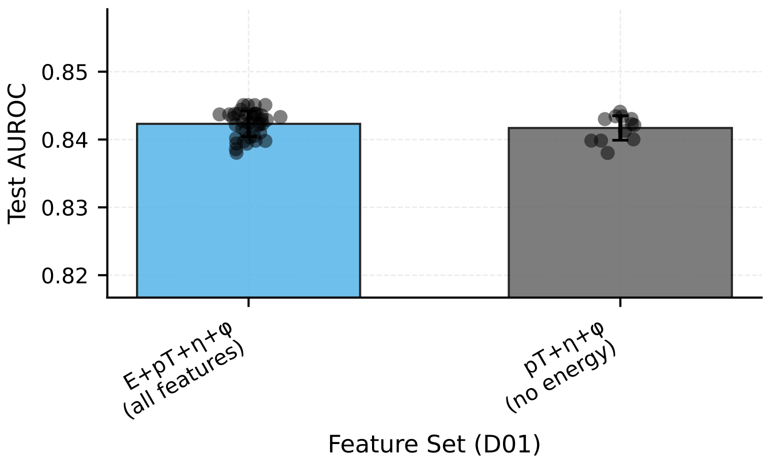

Figure 4.7: Mean test AUROC by feature set (D01) for the identity tokenizer. `[0,1,2,3]`: full four-vector (E, p_T, η, ϕ). `[1,2,3]`: three-vector (p_T, η, ϕ) without energy. The difference (-0.0007) is below the seed-spread level and constitutes a null result.

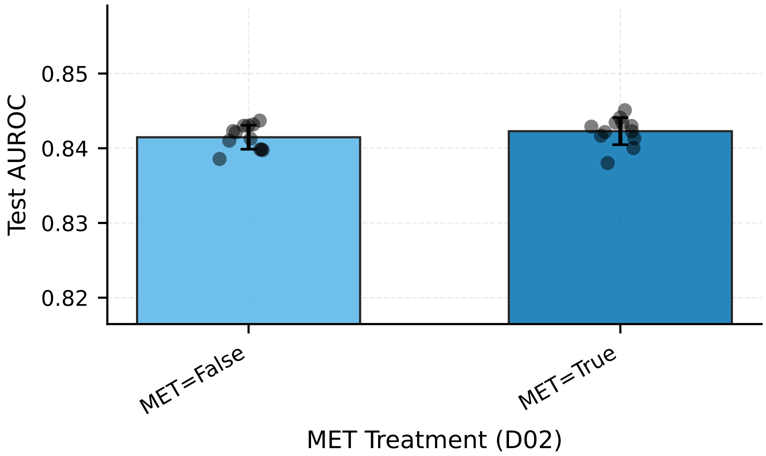

Figure 4.8: Mean test AUROC by MET treatment (D02), marginalised over D01 and D03 conditions. Including MET as a token yields a difference of $+0.0008$ AUROC, which is indistinguishable from noise at $n = 12$ per condition. This is a null result.

throughout the remainder of this thesis. Any chapter that modifies one of these axes re-runs the relevant comparison on top of this configuration.

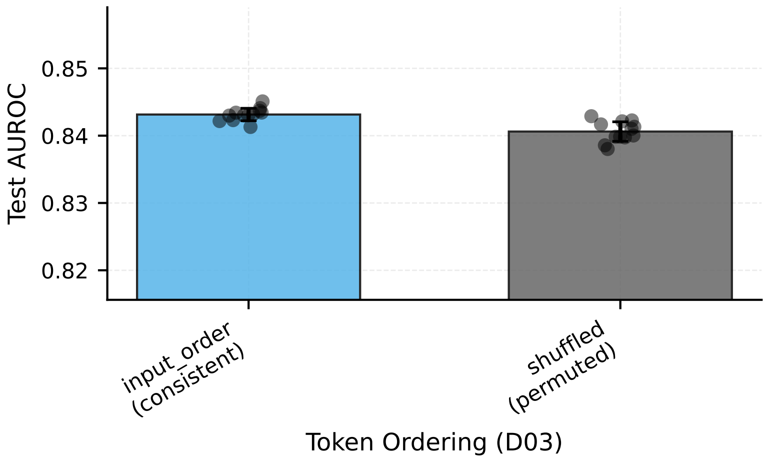

Figure 4.9: Mean test AUROC by token ordering (D03), for D02 = `false` conditions ($n = 6$ per bar). Shuffled token sequences yield -0.0026 AUROC relative to the original reconstruction order. The effect is consistent across feature-set conditions.

Axis	Config key	Chosen value	Justification
T1	`classifier.model.tokenizer.name`	`identity`	$+0.038$ AUROC over `raw`; dominant effect
T1-a	`classifier.model.tokenizer.pid_mode`	`one_hot`	robust at all dims; no trainable embedding weights
T1-b	`classifier.model.tokenizer.id_embed_dim`	`num_types`	set automatically by `one_hot`; no free param
D01	`data.cont_features`	`[0,1,2,3]`	null gap vs. `[1,2,3]`; required for P1 (node-wise mass)
D02	`classifier.globals.include_met`	`false`	null result; re-examined via B1-G1 in Ch. 6
D03	`data.sort_tokens_by`	`input_order`	small but consistent gain over `shuffled`

Table 4.4: Chapter 4 baseline configuration carried forward to all subsequent chapters. Each axis is held at the value shown unless explicitly varied.

5

Physics-Informed Attention Biases

Standard self-attention assigns relevance to token pairs through learned query–key inner products, making no assumption about the physical relationships among the particles those tokens represent. In a four-tops-versus-background event, however, pairs of particles that share a decay vertex, are separated by a small angle, or interact through a specific Standard Model (SM) coupling are physically more related than arbitrary token pairs. This chapter investigates whether encoding such structure as additive biases on the attention logits improves classification performance, and what the learned bias weights reveal about how the model organises particle interactions.

Four bias families are available in the workbench. The *Lorentz-scalar bias* (B1-L) maps pairwise kinematic invariants to attention scores through a small sub-network. The *type-pair table* (B1-T) is a learnable $N_{\text{type}} \times N_{\text{type}}$ matrix that assigns a baseline attention score to every ordered pair of particle types, optionally initialised from SM coupling constants. The *SM interaction prior* (B1-S) encodes SM coupling topology as a fixed or learned additive bias parametrised by coupling constants. The *global-conditioned bias* (B1-G) modulates attention using a global event-level feature such as missing transverse energy (MET) direction. All families share a zero-initialised scalar gate, so adding any bias leaves the transformer output unchanged at initialisation; the gate must be driven away from zero for the bias to have any effect. All experiments in this chapter hold D02 (`include_met`) at `true`: MET is always present, as motivated by the Ch. 4 findings, and is required for the B1-G family to operate.

The primary lens throughout this chapter is *interpretability*. Performance differences between bias configurations are expected to be small, and the main scientific contribution is understanding what structure the model learns to place on particle-pair relationships when given physics-motivated inductive biases. Section 5.1 establishes the performance envelope across all four families and examines attention maps as the central interpretability tool. Sections 5.2 through 5.4 drill into the internal design choices of each family, and Section 5.5 draws the chapter together.

5.1. Bias family comparison (Exp. 5A)

5.1.1. Experimental design

Experiment 5A asks whether any of the four bias families, applied individually or in full combination, measurably improves classification accuracy relative to an unbiased transformer baseline, and whether the full stack is super-additive. Eighteen runs cover six bias configurations at three seeds each: `none` (unbiased baseline), `lorentz_scalar`, `typepair_kinematic`, `sm_interaction`, `global_conditioned`, and all four combined. All other axes are held at the chapter baseline: embedding dimension 128 (H01), depth 3 (H02), 4 heads (H03), MLP dimension 256 (H04), tokeniser `identity` (T1), positional encoding `sinusoidal` (E1), and PID embedding mode `learned` with dimension 8.

5.1.2. Performance results

Table 5.2 reports seed-mean test AUROC and per-cell seed standard deviation ($n = 3$) for each configuration. The `none` baseline achieves 0.84125 ± 0.00068.

Axis	Values	Levels
B1 — Bias selector	none lorentz_scalar typepair_kinematic sm_interaction global_conditioned lorentz+typepair+sm+global	6
Seeds (R5)	42, 123, 314	3
Total		**18**

Table 5.1: Exp. 5A design. All other axes at chapter baseline: T1 = identity, D02 = true, E1 = sinusoidal, dim 128, depth 3, 4 heads.

B1 configuration	Mean AUROC	Seed std	Δ vs baseline
none (baseline)	0.84125	0.00068	—
lorentz_scalar	0.84156	0.00029	$+0.00031$
typepair_kinematic	0.84155	0.00080	$+0.00030$
sm_interaction	0.84080	0.00136	-0.00045
global_conditioned	0.84170	0.00091	$+0.00045$
lorentz+typepair+sm+global	0.83937	0.00090	-0.00188

Table 5.2: Exp. 5A seed-mean test AUROC ($n = 3$ seeds per cell). Per-cell seed standard deviation is indicative only; inter-cell range (≈ 0.002) is comparable to seed variance.

Three observations follow directly from these results. First, no single bias family moves the seed-mean AUROC by more than approximately one inter-seed standard deviation away from the none baseline. The three families that nominally improve accuracy (lorentz_scalar, typepair_kinematic, global_conditioned) each gain at most $+0.00045$ AUROC, which is smaller than the per-cell seed standard deviation of the baseline itself. Second, sm_interaction sits -0.00045 below baseline, within one seed standard deviation. Third, the full-combination cell is the only configuration whose mean (0.83937) sits visibly below the baseline, at a deficit of approximately 0.002 AUROC. This result is consistent with mild optimisation interference when multiple bias signals compete simultaneously for the same attention logit, though it cannot be cleanly attributed to any single component without further ablation.

The AUROC metric alone therefore does not reveal a meaningful benefit from physics-informed attention biases at this model scale and seed budget. The more informative question is whether the biases restructure attention towards physically motivated patterns even when the classification boundary shifts only modestly. The attention map analysis in Section 5.1.3 addresses this.

5.1.3. Attention structure: none versus full combination

Attention weights are extracted at inference time by forwarding 2000 validation events through each model with the capture_attention flag enabled, then averaging over all heads, all layers, and all three seeds. The physical token sub-block (18 particle tokens, positions 1–18 in the 21-token sequence; MET and MET-ϕ excluded) is projected onto a 7×7 type-pair grid by averaging attention from tokens of type i to tokens of type j for each ordered pair (i, j) across events. The seven physical types are: jet (1), b-jet (2), e^+ (3), e^- (4), μ^+ (5), μ^- (6), photon (7).

In the none baseline, type-pair attention weights occupy a near-uniform range of 0.059–0.113. No pair dominates: the largest cell ($\mu^+ \to e^-$: 0.113) and the smallest (inferred from the stated range) differ by only 0.054 units. The distribution is consistent with a model that has learned to aggregate broad context without privileging specific particle-type relationships.

The full-combination model produces a qualitatively different pattern. Table 5.3 lists the most strongly elevated and depressed type-pair cells. The attention range widens to 0.039–0.178, a factor of approximately 2.6 relative to the baseline range of 0.054. Critically, the elevated cells correspond to physically motivated interactions.

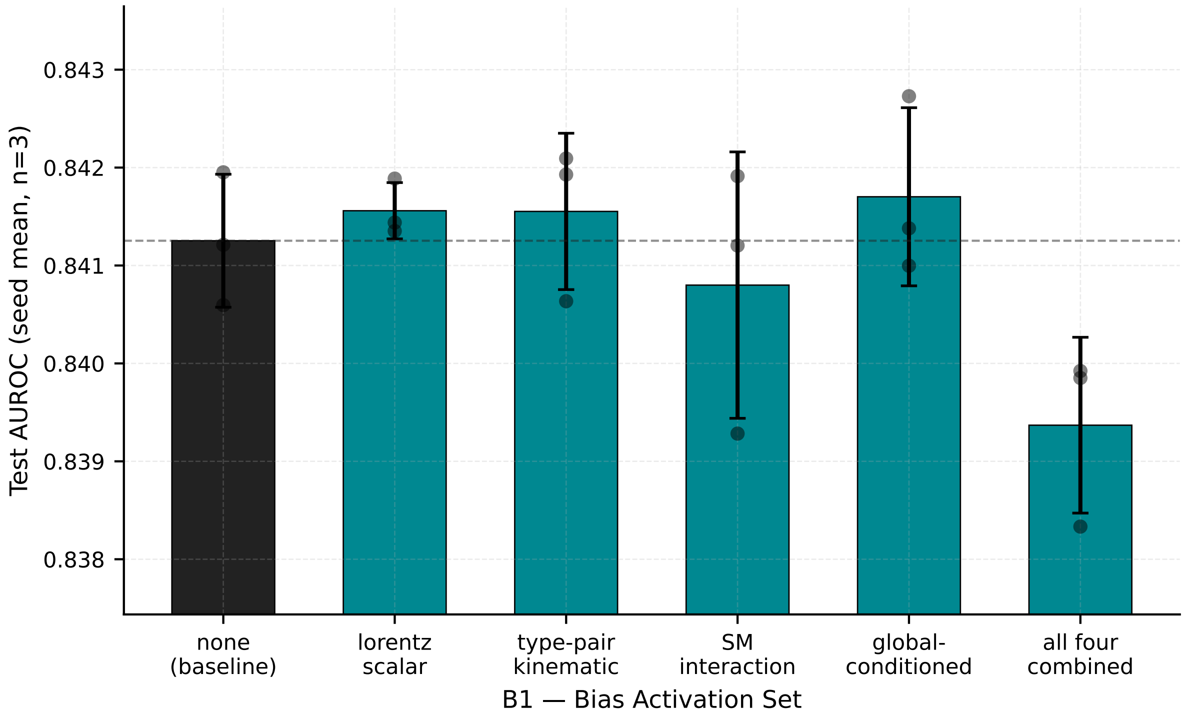

Figure 5.1: Exp. 5A seed-mean test AUROC for all six B1 configurations, with per-seed spread. Dashed line marks the `none` baseline mean.

The pattern is interpretable in HEP terms. Lepton–photon pairs ($\mu^- \to \gamma$: 0.178) are associated with final-state radiation or radiative W/Z decays. Photon self-attention (0.164) reflects the importance of photon isolation and cluster-shape information. The b-jet-to-muon cells (0.142, 0.133) directly encode the semi-leptonic b-decay topology relevant to top-quark reconstruction, where a b quark decays to a charm quark and a lepton. Opposite-sign lepton pairs ($e^- \to e^+$: 0.139, $e^+ \to e^-$: 0.129) are consistent with Z- or W-boson leptonic decays in the multi-top system. By contrast, same-charge mixed-lepton pairs ($\mu^- \to e^+$: 0.039) are strongly suppressed, as expected for pairs that share no direct SM vertex in the $t\bar{t}t\bar{t}$ decay chain.

These results constitute the primary interpretability finding of the chapter: the physics-informed biases do not substantially improve classification accuracy on this dataset at this model scale, but they do restructure the model's attention towards physically motivated particle-type relationships. The model, when given structured priors through the bias families, converges to an attention pattern that is broadly consistent with HEP domain knowledge, even though the AUROC improvement is within seed noise.

The per-head, per-layer attention grids (available in the supplementary figures) show that the structured pattern is not uniform across all twelve head–layer combinations. Some heads retain near-uniform attention while others specialise strongly onto physically motivated pairs, consistent with the expectation that multi-head attention allows functional specialisation.

5.2. Lorentz feature set and bias network (Exp. 5B)

5.2.1. Experimental design

The Lorentz-scalar bias sub-network (B1-L) computes a pairwise attention score from Lorentz-invariant quantities derived from pairs of four-momenta. Experiment 5B asks three questions about the internal design of this sub-network: which kinematic features are most informative (B1-L1), whether replacing the standard pointwise MLP with a Kolmogorov–Arnold Network (KAN) improves the learned response (B1-L2), and whether a per-feature sparse gate (B1-L5) can learn to prune uninformative features automatically. Sixty runs span five feature sets, two MLP types, and two gating settings, at three seeds each.

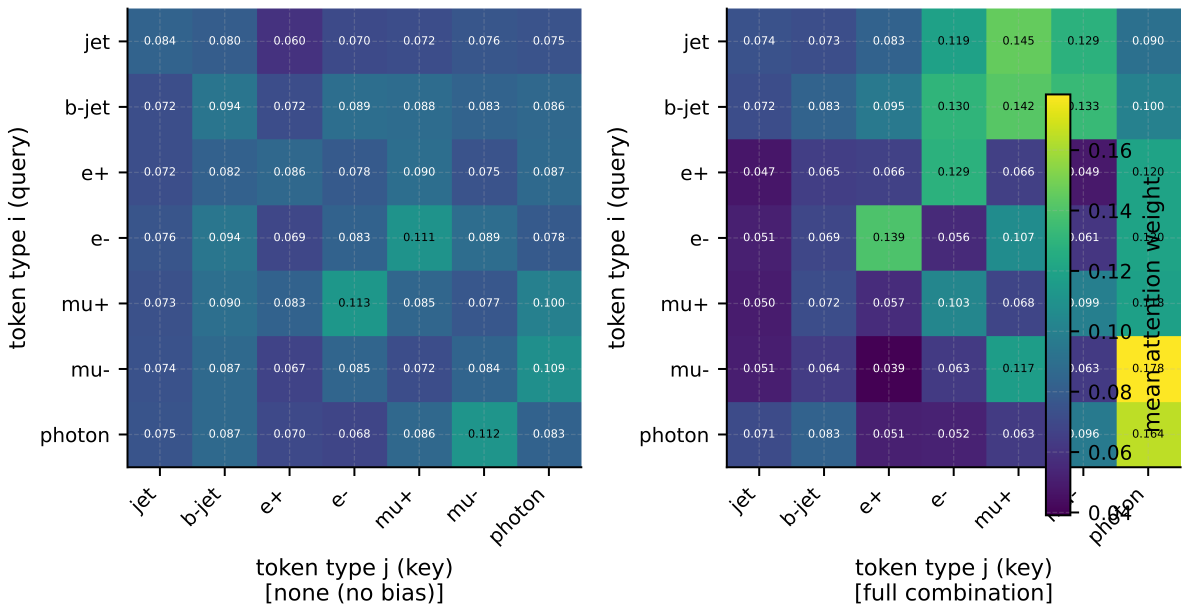

Figure 5.2: Type-pair mean attention weights for the `none` baseline (left) and full-combination model (right), averaged over all heads, layers, seeds, and 2000 validation events. Particle types: 1 = jet, 2 = b-jet, 3 = e^+, 4 = e^-, 5 = μ^+, 6 = μ^-, 7 = photon. Pad tokens excluded.

Pair $(i \to j)$	Mean attention	Physics interpretation
$\mu^- \to \gamma$	0.178	Lepton–photon (FSR / radiative return)
jet $\to \mu^+$	0.145	Jet-embedded muon (semi-leptonic b-decay)
$\gamma \to \gamma$	0.164	Photon self-attention (cluster isolation)
b-jet $\to \mu^+$	0.142	Direct $b \to \mu$ semi-leptonic
$e^- \to e^+$	0.139	Opposite-sign lepton pair (Z/W decay)
b-jet $\to e^-$	0.130	Semi-leptonic $b \to e$
$e^+ \to e^-$	0.129	Opposite-sign lepton pair
jet $\to \mu^-$	0.129	Jet-embedded muon
b-jet $\to \mu^-$	0.133	Semi-leptonic b-decay
$\mu^- \to e^+$	0.039	Same-charge mixed lepton (suppressed)
$e^+ \to \mu^-$	0.049	Same-charge mixed lepton (suppressed)

Table 5.3: Selected type-pair mean attention weights under the full-combination model (Exp. 5A), averaged over all heads, layers, seeds, and 2000 validation events. Pairs with mean attention ≥ 0.13 are elevated; pairs ≤ 0.05 are depressed.

All kinematic features are computed from z-score-normalised four-vectors. An important consequence of this normalisation is that the pairwise invariant mass m^2 collapses near zero on the normalised scale (validation median of m^2 is 0.000; 95th percentile is 2.47). The logarithmic variant $\log m^2$ collapses further (validation median $\approx \log(10^{-6}) \approx -13.8$). This means that any claim about "learning the W-boson or top-quark mass peak" is not supported by these checkpoints; such an analysis would require either removing input normalisation or de-normalising the feature axis. By contrast, ΔR, $\log k_T$, and z remain well-behaved on the normalised scale (ΔR at the 5th and 95th percentiles: 0.79 and 4.17 respectively), and interpretability arguments based on those features are on firmer ground.

5.2.2. AUROC results

Table 5.5 reports seed-mean AUROC for all 20 (feature set, MLP type, gating) cells. The full bar chart is shown as Figure 5.4.

Three consistent trends emerge from the table. First, within every (feature-set, gating) pairing, the `kan` MLP equals or exceeds the `standard` MLP in seed-mean AUROC. The largest gap appears in the six-feature, gating-off configuration ($+0.0018$ AUROC), which equals approximately one seed standard

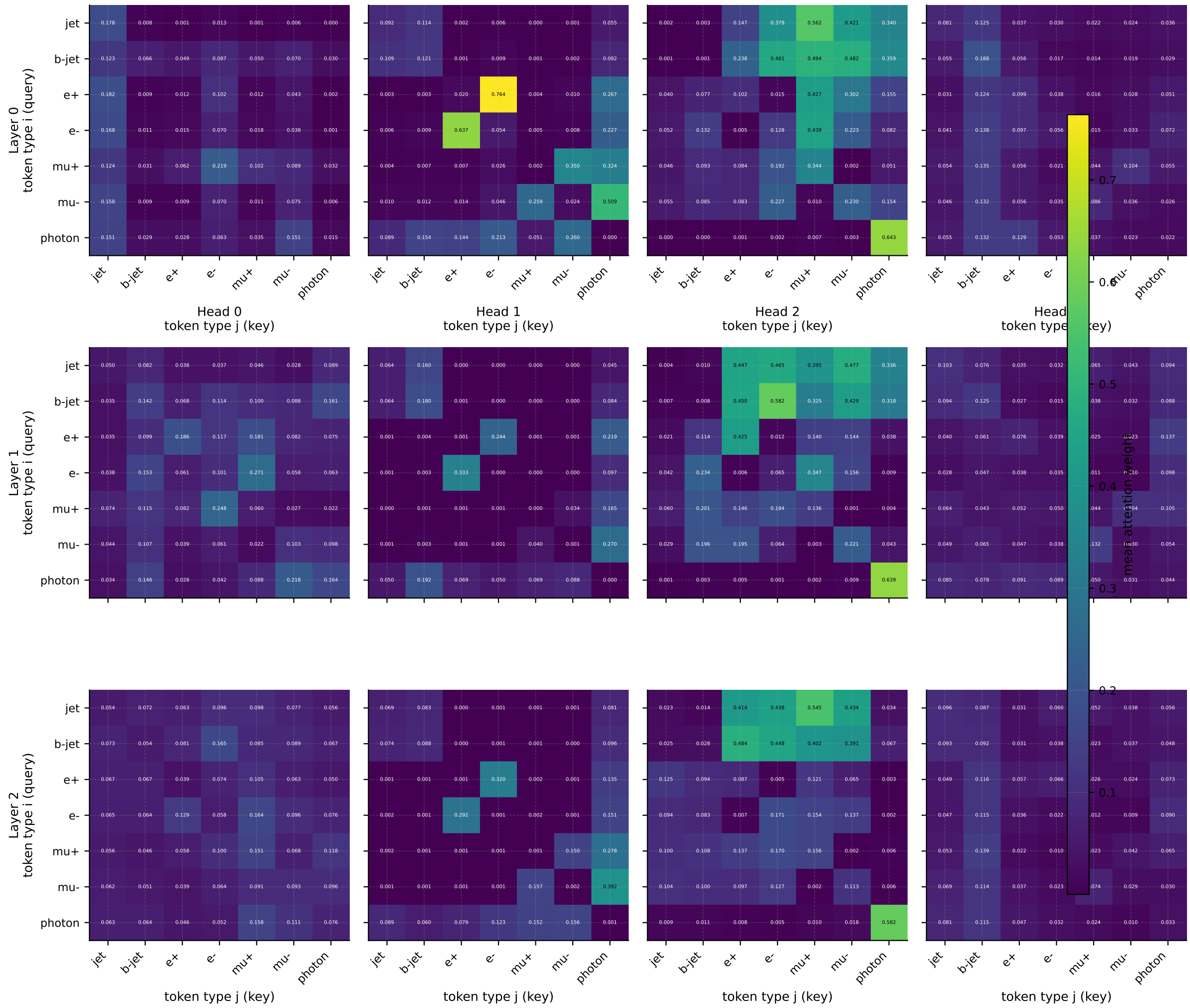

Figure 5.3: Type-pair attention weights for the full-combination model (seed 42), shown separately for each of the 4 heads × 3 encoder layers. Some heads specialise onto physically motivated pairs while others retain near-uniform attention.

deviation. The global best cell (6-feature, KAN, gate off: 0.84436) sits roughly $+0.0031$ above the Exp. 5A none baseline (0.84125), a margin comparable to several per-cell seed standard deviations. Second, richer Lorentz feature panels systematically produce higher AUROC than single-feature panels, particularly when paired with the KAN. The four- and six-feature sets consistently outperform the ΔR-only and m^2-only single-feature sets; this trend is visible in both the standard and KAN columns. Third, sparse gating provides no benefit at this scale and seed budget: gate-on cells tend to sit slightly below their gate-off counterparts in the same (feature, MLP) row, by approximately 0.0005–0.0009 AUROC.

5.2.3. KAN spline structure and gate values

The expressive difference between KAN and standard MLP is visible not only in AUROC but in the learned attention-bias functions themselves. In the single-feature (ΔR) configuration with gating off, the seed-mean bias function spans a range of 0.99 arbitrary units under the KAN, versus 0.46 under the standard MLP: a factor of approximately two. In the partial-dependence view of the four-feature configuration (other features held at validation medians), the standard MLP learns almost no log-k_T dependence (range ≈ 0.05), while the KAN learns a structured curve with range ≈ 0.65. This result is consistent with the KAN's capacity to model sharp univariate edges with a small parameter budget through its B-spline basis. The KAN therefore extracts more functional structure from the pairwise kinematic features, which is reflected in the consistent AUROC advantage.

The per-module gate, $\tanh(g)$, converged to a positive value in every one of the 60 Exp. 5B runs, confirming that the Lorentz bias is non-trivially active throughout the cohort. KAN modules average

Axis	Values	Levels

B1-L1 — Feature set	`[deltaR]`
	`[m2]`
	`[m2, deltaR]`
	`[log_kt, z, deltaR, log_m2]`
	`[m2, deltaR, log_m2, log_kt, z, deltaR_ptw]`
		5

B1-L2 — MLP type	`standard`, `kan`	2
B1-L5 — Sparse gating	`false`, `true`	2
Seeds (R5)	42, 123, 314	3
Total		**60**

Table 5.4: Exp. 5B design. B1 = `lorentz_scalar` throughout. B1-L3 (hidden dim) and B1-L4 (per-head mode) are held at defaults.

Feature set	MLP type	Gating	Mean AUROC	Seed std
ΔR	standard	off	0.84126	0.00123
ΔR	kan	off	0.84245	0.00089
ΔR	standard	on	0.84155	0.00126
ΔR	kan	on	0.84206	0.00044
m^2	standard	off	0.84131	0.00142
m^2	kan	off	0.84154	0.00159
m^2	standard	on	0.84114	0.00118
m^2	kan	on	0.84117	0.00118
m^2, ΔR	standard	off	0.84114	0.00042
m^2, ΔR	kan	off	0.84256	0.00063
m^2, ΔR	standard	on	0.84100	0.00104
m^2, ΔR	kan	on	0.84190	0.00022
$\log k_T$, z, ΔR, $\log m^2$	standard	off	0.84200	0.00085
$\log k_T$, z, ΔR, $\log m^2$	kan	off	0.84399	0.00101
$\log k_T$, z, ΔR, $\log m^2$	standard	on	0.84191	0.00034
$\log k_T$, z, ΔR, $\log m^2$	kan	on	0.84311	0.00076
m^2, ΔR, $\log m^2$, $\log k_T$, z, $\Delta R/p_T$	standard	off	0.84257	0.00106
m^2, ΔR, $\log m^2$, $\log k_T$, z, $\Delta R/p_T$	kan	off	**0.84436**	0.00214
m^2, ΔR, $\log m^2$, $\log k_T$, z, $\Delta R/p_T$	standard	on	0.84170	0.00065
m^2, ΔR, $\log m^2$, $\log k_T$, z, $\Delta R/p_T$	kan	on	0.84378	0.00107

Table 5.5: Exp. 5B seed-mean test AUROC ($n = 3$ per cell). The `none` baseline from Exp. 5A is 0.84125 ± 0.00068. Best cell highlighted. Gating "on" = B1-L5 = `true`.

$\tanh(g) \approx 0.28 \pm 0.08$ versus 0.22 ± 0.10 for standard modules. The higher gate for KAN runs echoes the AUROC finding: the network allows more of the KAN's richer signal through its multiplicative gate.

Despite the gate being consistently open, the sparse feature-gating mechanism (B1-L5) does not learn meaningful per-feature selection. The sigmoid gate values $\sigma(\text{feature\_gates})$ remain within 0.49–0.62 across all 15 sparse-on KAN runs, barely departing from their initialisation at $\sigma(0) = 0.5$. The only feature whose mean gate drifts slightly below 0.5 is $\log m^2$, which is precisely the feature whose input distribution collapses near -13.8 under z-score normalisation. Even this drift is modest ($\approx 0.494 \pm 0.022$), and no feature is pruned in any run. The effect on AUROC is therefore a slight attenuation of every feature's amplitude simultaneously, with no compensating benefit from pruning, which explains the small but systematic gap between gate-on and gate-off cells.

One caveat for the spline-shape analysis is that the KAN B-spline grid is fixed at its initialisation throughout training: `update_grid` is not called during the forward pass, which was confirmed by state-dict inspection of the `mlp.0.grid` key in the saved checkpoints. The spline expressivity is therefore bounded by the initial grid resolution and not by adaptive grid relocation. More adaptive grid training could in principle widen the gap further.

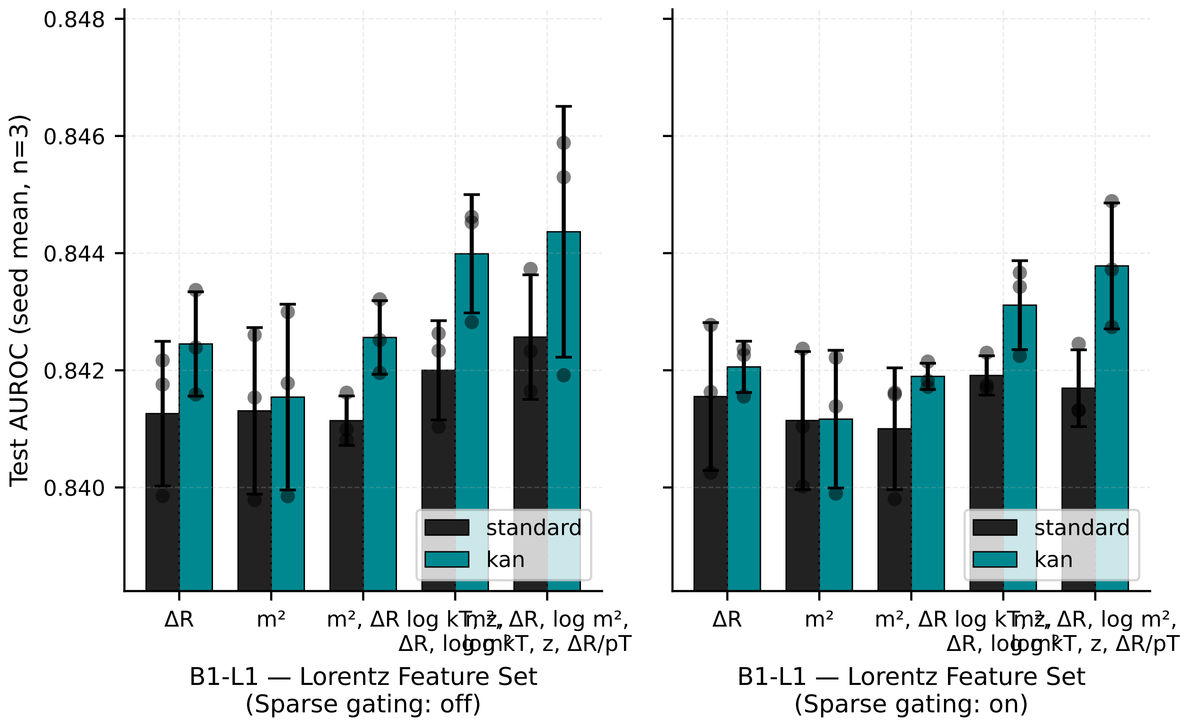

Figure 5.4: Exp. 5B seed-mean test AUROC for all 20 (feature set, MLP type, gating) cells, with per-seed spread. KAN consistently meets or exceeds the standard MLP; richer feature sets produce higher AUROC; sparse gating provides no benefit.

5.3. Type-pair initialisation and freeze (Exp. 5C)

5.3.1. Experimental design

The type-pair kinematic bias (B1-T) maintains a learnable $N_{\text{type}} \times N_{\text{type}}$ table of attention score offsets, one per ordered particle-type pair. Experiment 5C varies two design choices: the initialisation strategy (B1-T1) and whether the table is frozen after initialisation (B1-T2). Three initialisation modes are tested: `none` (zero start, free to train), `binary` (a $\{0, -5\}$ mask encoding which SM vertices are allowed or forbidden), and `fixed_coupling` (the physics-motivated initialisation, with magnitudes proportional to SM coupling constants). A frozen table (`freeze_table = true`) acts as a fixed physics prior; a free table is adjusted by gradient descent from its initialised state. Fifteen runs cover five cells (the `none`–frozen combination is excluded, as there is no table to freeze) at three seeds each.

Axis	Values	Levels
B1-T1 — Init	`none`, `binary`, `fixed_coupling`	3
B1-T2 — Freeze	`false`, `true` (only when init $\neq$ `none`)	—
Seeds (R5)	42, 123, 314	3
Total		**15**

Table 5.6: Exp. 5C design. B1 = `typepair_kinematic` throughout. B1-T3 (kinematic gate), B1-T4 (kinematic feature), and B1-T5 (mask value) are held at defaults.

5.3.2. Performance results

Table 5.7 reports seed-mean AUROC for the five cells. The local `none`-init baseline here (0.84205) is slightly above the Exp. 5A `none` cell (0.84125), a difference attributable to the different sweep configuration and cohort, not to a change in the architecture.

All four physics-initialised cells sit below the `none` baseline in seed mean, by margins of 0.0003–0.0013 AUROC. These deltas are within one to two per-cell seed standard deviations and should not be interpreted as a meaningful performance deficit. The `fixed_coupling` initialisation consistently outperforms

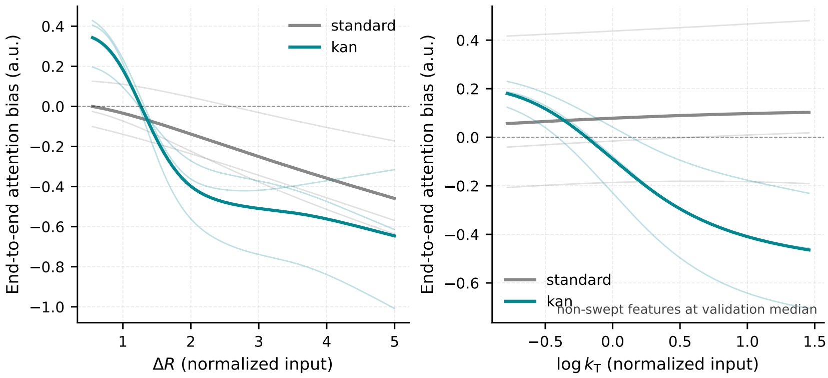

Figure 5.5: Learned 1D bias function for the ΔR-only feature set comparing KAN (blue) and standard MLP (orange), averaged over three seeds. The KAN spans a range of ≈ 0.99 attention units versus ≈ 0.46 for the standard MLP.

Init	Freeze	Mean AUROC	Seed std	Δ vs `none`
`none`	free	0.84205	0.00081	—
`binary`	free	0.84075	0.00046	-0.00130
`binary`	frozen	0.84131	0.00065	-0.00074
`fixed_coupling`	free	0.84175	0.00102	-0.00030
`fixed_coupling`	frozen	0.84138	0.00046	-0.00067

Table 5.7: Exp. 5C seed-mean test AUROC ($n = 3$ per cell). All four physics-initialised cells sit below the `none` baseline within one to two seed standard deviations.

`binary` by approximately 0.0006–0.0010 AUROC regardless of freeze state, but again at the noise floor. Freezing the table has small, non-monotonic effects: it slightly improves `binary` ($+0.00056$) and slightly reduces `fixed_coupling` (-0.00037), both within seed variance.

The AUROC results therefore suggest neither a benefit from SM-seeded initialisation nor a cost from freezing the physics prior, at this seed budget and model scale.

5.3.3. Learned table structure

The more informative view comes from inspecting the converged type-pair tables directly. The state-dict key `bias_composer.bias_modules.typepair_kinematic.table_raw` has shape $[8, 8]$; the symmetric form is reconstructed as $\frac{1}{2}(T + T^{\top})$ and the padding row and column are removed, leaving a 7×7 active matrix. Table 5.8 summarises key scalar statistics.

Init	Freeze	Gate ($\tanh g$, mean $\pm$ std)	Frobenius norm	Drift from init
`none`	free	0.234 ± 0.015	0.434	0.944 (= norm)
`binary`	free	0.097 ± 0.012	27.044	0.868
`binary`	frozen	0.094 ± 0.013	27.295	0.000 (frozen)
`fixed_coupling`	free	0.104 ± 0.015	26.848	0.930
`fixed_coupling`	frozen	0.101 ± 0.015	27.094	≈ 0

Table 5.8: Exp. 5C per-group learned-table statistics. Gate = $\tanh(g)$ averaged over 3 seeds. Frobenius norm computed on the 7×7 active sub-matrix. Drift from init = Frobenius distance between the converged seed-mean table and the reference initialisation.

The `none`-init table starts from zero and converges to a small-magnitude matrix (Frobenius 0.43), with values scattered in $[-0.16, +0.12]$ showing no discernible SM-interaction pattern at $n = 3$ seeds. Its gate is notably higher (0.234) than any physics-initialised group (0.094–0.104), consistent with the network opening the gate further when the table provides a weak prior and all type pairs look equally neutral at initialisation.

The `fixed_coupling`-free table preserves the SM structure through training. QCD quark–quark pairs (jet–jet: $+1.22$, jet–b-jet: $+1.17$, b-jet–b-jet: $+1.23$) retain large positive values; EW lepton pairs (e^+e^-:

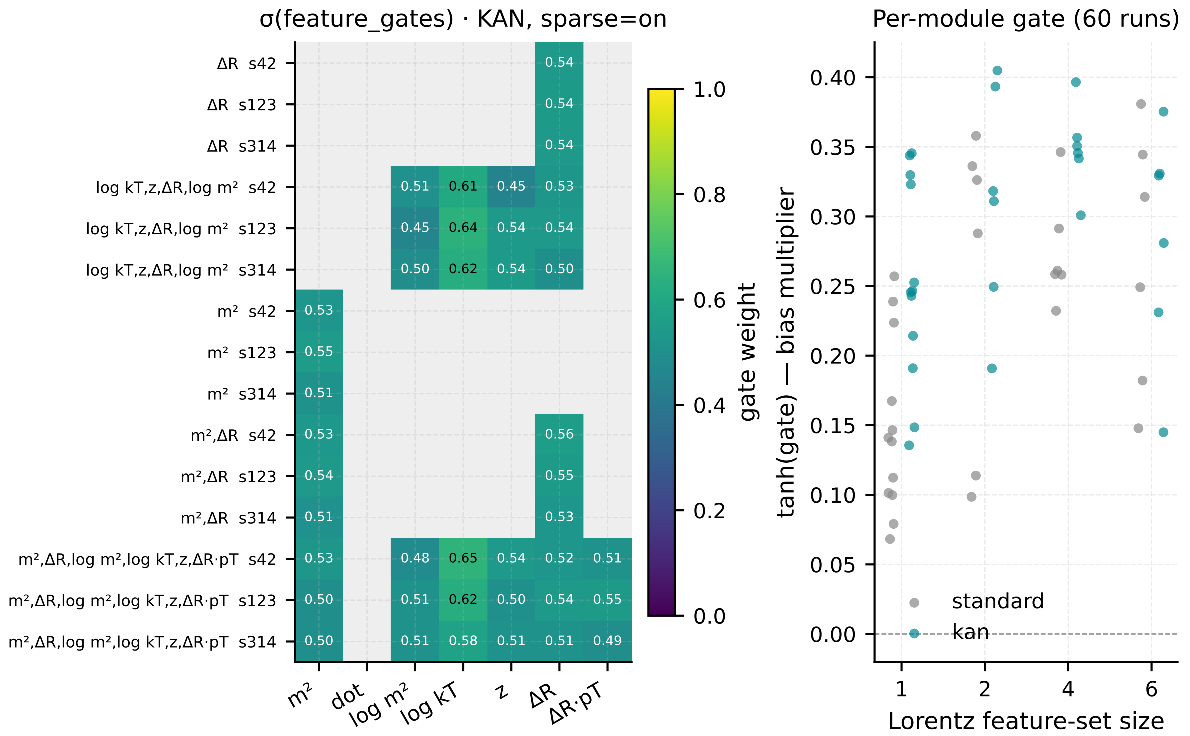

Figure 5.6: Left: per-feature sparse gate values $\sigma(\text{gate})$ for all Exp. 5B sparse-on KAN runs; all gates remain within 0.49–0.62, indicating no meaningful feature pruning. Right: module-level scalar gate $\tanh(g)$ for KAN (blue) and standard (orange) MLP runs across the full cohort.

$+0.54$, $\mu^+\mu^-$: $+0.89$) retain intermediate positive values; and all non-interacting pairs remain near the mask value of -5.0. The Frobenius drift from initialisation is 0.930, indicating that training adjusts the coupling-constant magnitudes by roughly ± 0.1–0.3 rather than restructuring the topology. This is a notable result: the network broadly respects the SM interaction hierarchy encoded at initialisation, even when the gradient is free to reorganise the table entirely.

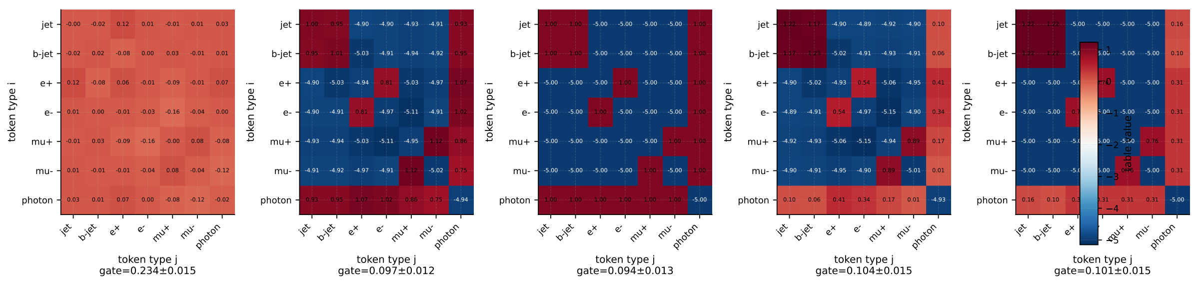

Figure 5.7: Converged type-pair tables (7×7 active sub-matrix) for all five Exp. 5C groups, averaged over three seeds. The `fixed_coupling`-free table preserves the SM coupling hierarchy; the `none`-init table remains near-uniform.

The `binary`-free group shows an analogous pattern (same mask positions preserved) but with higher drift (0.868 vs 0.930), reflecting the fact that the binary initialisation provides structural topology but no coupling-constant scale, leaving active entries more exposed to gradient learning. The frozen groups confirm the mechanism functions correctly (freeze drift $\approx 6.6 \times 10^{-8}$ for `fixed_coupling`-frozen).

Taken together, the results suggest that the SM `fixed_coupling` initialisation provides a structural prior that training broadly preserves, but that its effective attention contribution, modulated by the scalar gate ($\tanh(g) \approx 0.10$), is too small to produce a measurable AUROC improvement at this configuration. The `none`-init group, starting from a flat table, fails to recover a physics-consistent structure at three seeds — the table remains near-uniform, indicating that the inductive bias from the initialisation is important for guiding the table to a structured solution.

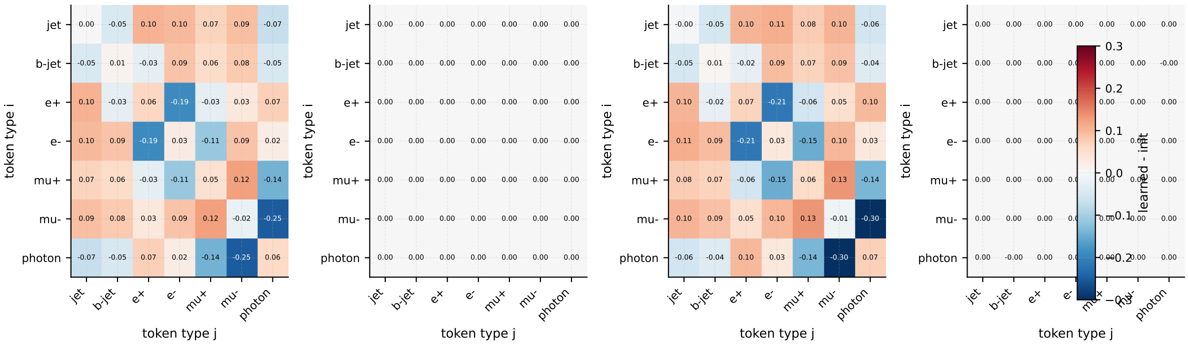

Figure 5.8: Per-cell table drift from initialisation for the free-table groups. `fixed_coupling`-free (top) shows small, structured adjustments (± 0.1–0.3) with the SM topology intact; `binary`-free (bottom) shows larger drift at active entries.

5.4. SM interaction mode (Exp. 5D)

Experiment 5D tests whether increasing physical fidelity in the SM interaction prior produces measurable gains. The `sm_interaction` bias family encodes SM topology as a fixed or modulated additive bias on attention logits; axis B1-S1 selects among three modes: `binary` (a $\{0, -5\}$ allowed/forbidden mask), `fixed_coupling` (coupling constants at a fixed energy scale), and `running_coupling` ($\alpha_s(Q^2)$ evaluated at the pairwise momentum transfer). Nine runs cover all three modes at three seeds each.

Axis	Values	Levels
B1-S1 — SM mode	`binary`, `fixed_coupling`, `running_coupling`	3
Seeds (R5)	42, 123, 314	3
Total		**9**

Table 5.9: Exp. 5D design. B1 = `sm_interaction` throughout. B1-S2 (mask value) held at default.

Table 5.10 reports seed-mean AUROC. The reference `none` baseline from Exp. 5A (0.84125 ± 0.00068) is used for comparison, as Exp. 5D does not include an explicit `none` cell; both cohorts share the same encoder hyperparameters and seed list, so the cross-cohort comparison is reasonable.

B1-S1 mode	Mean AUROC	Seed std	Δ vs 5A `none`
`binary`	0.84140	0.00050	$+0.00015$
`fixed_coupling`	0.84146	0.00091	$+0.00021$
`running_coupling`	0.84108	0.00081	-0.00018

Table 5.10: Exp. 5D seed-mean test AUROC ($n = 3$ per cell). All three modes are within ± 0.0002 AUROC of the Exp. 5A `none` baseline.

All three SM-interaction modes sit within ± 0.0002 AUROC of the `none` baseline, and the three modes span only ≈ 0.0004 AUROC among themselves. Both magnitudes are at or below the per-cell seed standard deviation, making no mode distinguishable from another within this seed budget. The nominal ranking (`fixed_coupling` $>$ `binary` $>$ `running_coupling`) is consistent with the `running_coupling` mode being the most parameter-rich and therefore most exposed to optimisation noise at fixed hyperparameters, but this interpretation should not be treated as a directional finding at $n = 3$ seeds.

The SM interaction prior as implemented in Exp. 5D is therefore a null result on the AUROC axis at this model scale. Whether the prior is active in the attention pattern is a question for a future attention-map follow-up analogous to the Exp. 5A analysis (currently staged but not yet run). For the purpose of the downstream best-model selection in Ch. 8, the `fixed_coupling` mode is carried forward as the default SM setting on the grounds of physical motivation, accepting that the performance difference is negligible.

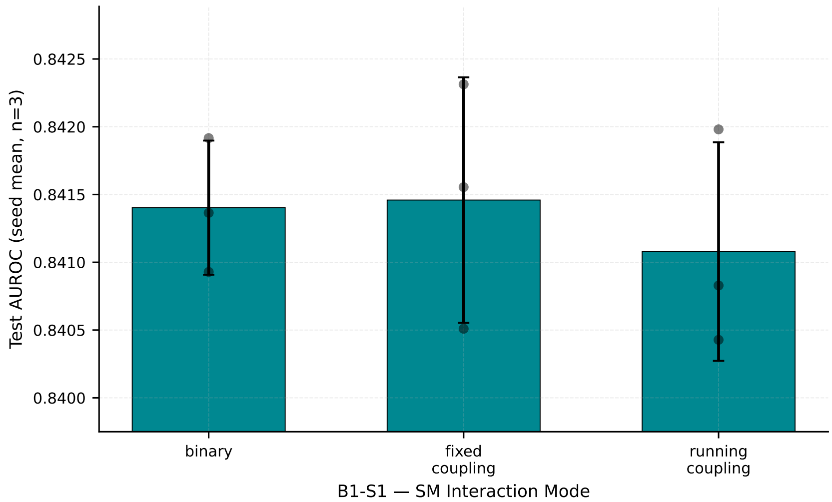

Figure 5.9: Exp. 5D seed-mean test AUROC for the three B1-S1 modes, with per-seed spread. Dashed line marks the Exp. 5A none baseline. All three modes are within ± 0.0002 AUROC of the baseline.

5.5. Chapter synthesis

This chapter set out to ask whether physics-informed attention biases improve 4-tops-versus-background classification, and what those biases reveal about the model's internal organisation of particle relationships. The AUROC results across all four sub-experiments are consistent and unambiguous: no single bias family, and no configuration within any family, produces a performance gain that is clearly separable from seed variance at $n = 3$ seeds per cell. The total AUROC range across all 102 runs is approximately 0.838–0.846, with per-cell seed standard deviations of order 0.001. The full-combination configuration in Exp. 5A is the only cell that sits visibly below the none baseline (≈ -0.002 AUROC), suggestive of optimisation interference from multiple simultaneous bias signals.

The interpretability results tell a richer story. When all four bias families are active, the model sharpens type-pair attention onto physically motivated particle relationships: the attention range widens by a factor of 2.6 (from 0.054 in the none baseline to 0.139 in the full-combination model), and the most elevated pairs are lepton–photon ($\mu^- \to \gamma$: 0.178), photon self-attention (0.164), b-jet-to-muon pairs consistent with semi-leptonic b-decay (0.142, 0.133), and opposite-sign lepton pairs consistent with Z/W decay ($e^- \to e^+$: 0.139). Same-charge mixed-lepton pairs, which lack a common SM vertex, are strongly suppressed. These patterns demonstrate that physics-informed biases restructure the model's attention rather than merely adding capacity.

Within the Lorentz-scalar family, KAN sub-networks consistently learn richer bias functions than standard MLPs across the same feature sets, with the largest gap at the six-feature configuration ($+0.0018$ AUROC, bias-function range $2\times$ wider). Sparse feature gating, however, converges to the all-open state ($\sigma \approx 0.5$) in every run, providing no automatic feature selection at this training budget.

The type-pair table analysis shows that the SM `fixed_coupling` initialisation provides a structural prior that training broadly respects: QCD quark–quark and EW lepton-pair entries retain their physics-consistent magnitudes through gradient descent, with Frobenius drift 0.930 from the initialisation. By contrast, a zero-start (`none`-init) table fails to recover a physics-consistent structure at $n = 3$ seeds, suggesting that the inductive bias from a physics-motivated initialisation is beneficial for interpretability even when it does not shift accuracy.

The practical takeaway for the remainder of the thesis is that physics-informed attention biases do not change the operating point of the classifier at the model scale studied in this chapter. Their value is interpretive: they make the model's attention patterns legible in HEP terms. For downstream experiments in Ch. 6 and Ch. 8, the chapter baseline carries forward the `lorentz_scalar` bias in the six-feature, standard-MLP, gating-off configuration as the default B1 setting, on the grounds that it provides the most consistent, modest AUROC benefit without the optimisation risk of the full-combination stack.

Experiment	Primary axes	Runs
5A — Family comparison	B1	18
5B — Lorentz ablation	B1-L1, B1-L2, B1-L5	60
5C — Type-pair table	B1-T1, B1-T2	15
5D — SM interaction mode	B1-S1	9
Total		**102**

Table 5.11: Chapter 5 run summary.

6
Attention Mechanisms

The attention mechanism is the only component that operates over all token pairs simultaneously. Chapters 4 and 5 asked how to enrich the information arriving at the attention layer; this chapter asks whether the attention computation itself should be modified. The central question is whether differential attention,which computes the difference of two softmax maps to suppress diffuse, low-information attention patterns, transfers from its original NLP setting to particle classification.

The physics motivation is direct: in a multi-top event, only a small subset of the 10–20 reconstructed objects are decay products of the four top quarks. Standard softmax attention distributes probability mass across all token pairs including unrelated soft objects. Differential attention is designed for exactly this setting, and provides a clear testable hypothesis.

Two experiments are run. Exp. 6A establishes the headline comparison across attention type, internal normalisation, and the presence of a Lorentz-scalar bias. Exp. 6B is a targeted follow-up on how physics biases interact with the two differential attention branches.

6.1. Attention type and internal normalisation (Exp. 6A)

Lens: performance and efficiency.

Sweeps A3 (standard vs. differential), A4 (attention-internal normalisation), and whether a Lorentz-scalar bias is active. The bias inclusion tests whether physics structure in the attention logits and noise suppression from differential attention are complementary or redundant. Wall-clock time per epoch is recorded alongside AUROC, since differential attention runs two softmax passes per head.

Axis	Values	Config
A3 — Attention type	`standard`, `differential`	2
A4 — Internal norm	`none`, `layernorm`, `rmsnorm`	3
B1 — Lorentz bias	`none`, `lorentz_scalar`	2
Seeds		3
Total		**36**

Table 6.1: Exp. 6A. All other axes at chapter baseline. `lorentz_scalar` uses the best feature set from Ch. 5.

Outputs: seed-averaged AUROC bar chart for all 12 configurations, with wall-clock time as a secondary axis. Attention maps extracted at inference time for standard and differential (both with A4=none, B1=none), averaged over a fixed held-out batch and labelled by particle type. If differential attention is working as intended, maps should show higher concentration on physically proximate pairs.

6.2. Differential bias mode (Exp. 6B)

Lens: performance.

Restricted to A3=`differential`. Tests how the additive Lorentz-scalar bias from Ch. 5 should interact with the two attention branches: added to neither, shared across both, or split with separate parameters per branch.

Axis	Values	Config
A3-a — Diff bias mode	`none`, `shared`, `split`	3
Seeds		3
Total		**9**

Table 6.2: Exp. 6B. A3=`differential` and B1=`lorentz_scalar` fixed throughout.

Output: compact three-row AUROC table. If `shared` and `split` perform equivalently, the second branch does not require independent bias information and the simpler shared parameterisation is sufficient.

Chapter summary

Experiment	Primary axes	Runs
6A — Attention type and norm	A3, A4, B1	36
6B — Differential bias mode	A3-a	9
Total		**45**

7
Model Scaling and Efficiency

Every architectural finding in Chapters 4–7 was established at a fixed model size. This chapter asks whether those findings hold at other scales, and where on the performance-cost curve a practitioner should operate. A secondary question is whether scaling behaviour depends on task difficulty: a 1v1 discrimination problem and a multiclass problem may saturate at different scales for physically meaningful reasons.

All experiments use a common fixed baseline configuration: T1 = `identity`, best D01 from Ch. 4, E1 = `sinusoidal`, no physics biases. This isolates scaling as the sole variable. Model capacity is controlled via the convenience key `axes/model_size_key`, which bundles dim, depth, heads, and FFN dim into named presets spanning roughly 50k to 4M parameters.

7.1. Scaling curves across tasks (Exp. 8A)

Lens: performance and efficiency.

Trains all four size presets on three classification tasks. The central question is whether the saturation point, the model size beyond which additional parameters yield diminishing returns, shifts with task difficulty. A steeper and later-saturating curve on the harder 1v1 tasks would reflect a genuine physics difference, not a modelling artefact.

Axis	Values	Config
Model size	`d32_L3`, `d64_L6` `d128_L8`, `d256_L12`	4
Task	4t vs. background 4t vs. ttH multiclass (all five processes)	3
Seeds		3
Total		**36**

Table 7.1: Exp. 8A. Common baseline configuration throughout.

Outputs: three scaling curves (one per task) on the same axes, showing seed-averaged AUROC vs. parameter count on a log scale. A secondary panel shows signal efficiency at fixed background rejection $1/B = 50$ as a physics-operational complement to AUROC.

7.2. Efficiency metrics (Exp. 8B)

No new training required. For each checkpoint from Exp. 8A, three efficiency metrics are recorded: inference throughput in events per second, training wall-clock time per epoch, and epochs to 95% of final AUROC. These are plotted as a Pareto panel against AUROC, one curve per task. The model sitting on the Pareto frontier at each task is the practical recommendation for that setting.

7.3. Axis transfer across scales (Exp. 8C)

Primary task only.

A targeted check that the axis rankings established in Chapters 4–7 are stable across scales. The two most impactful axes from Ch. 8 are trained at the smallest and largest presets in addition to the default, using the primary task only. Scale-dependent conclusions would imply the Part II findings are specific to the medium-size regime; scale-independent conclusions support them as broadly applicable.

Axis	Values	Config
Top-2 axes from Ch. 8	best vs. default setting	2
Model size	`d32_L3`, `d64_L6`, `d256_L12`	3
Seeds		3
Total		**18**

Table 7.2: Exp. 8C. Primary task (4t vs. background) only. Top-2 axes determined after Ch. 8 analysis is complete.

Chapter summary

Experiment	Description	Runs
8A — Scaling curves	4 sizes $\times$ 3 tasks $\times$ 3 seeds	36
8B — Efficiency	post-hoc from 8A checkpoints	0
8C — Axis transfer	2 axes $\times$ 3 sizes $\times$ 3 seeds	18
Total		**54**

Part III

Synthesis

8

What Matters: A Global Analysis

The preceding chapters each tested a specific axis group in isolation. This chapter inverts that framing and treats the entire experiment database, all models trained across Chapters 4–6 plus the existing exploratory runs, as the object of study. The question is global: across everything trained, which architectural axes consistently move the needle, and which have no measurable effect?

No new models are trained here. At the end of all training the W&B database is frozen, all runs are backfilled with a standardised axis vector using the V2 reference, and the result is a single clean table: one row per model, one column per axis, plus AUROC and efficiency metrics. All analysis in this chapter operates on that table.

8.1. The run database

After backfilling, every run carries a complete axis vector covering T1, T1-a, T1-b, D01–D03, E1, A1–A6, F1, B1 and its active sub-axes, C1, and the model-size keys H01–H04. Runs are filtered to the primary task (4t vs. combined background) and the default model size to keep AUROC values on a comparable scale. The resulting table contains roughly N models and forms the sole data source for this chapter.

8.2. Marginal sensitivity analysis

For each axis, the conditional distribution $P(\text{AUROC} \mid \text{axis} = s_i)$ is estimated by grouping all runs sharing setting s_i and computing the group mean, marginalising over all other axes. Because sweeps are approximately orthogonal, these marginal means are interpretable as main effects. The primary summary statistic per axis is the range: the difference between the highest and lowest group mean. Axes where this range falls below the seed-level noise floor ($\lesssim 0.002$ AUROC) are classified as axes that do not matter.

The main output is a single ranked bar chart spanning all axes, colour-coded by family (D, T, E, A, F, B, C). This is the headline figure of the chapter. Selected null results, including PE type, norm policy, and causal masking, are each given a small panel showing the flat marginal distribution, making the null result explicit rather than just asserted.

8.3. Surrogate modelling and configuration prediction

Beyond marginal effects, a gradient-boosted decision tree is trained on the run database to predict AUROC from the full axis configuration vector. This captures non-linear interactions between axes that the marginal analysis cannot detect, for example a bias family that only helps when combined with the identity tokeniser.

The idea is simple: treat every trained model as a data point, with its axis settings as features and its AUROC as the label. Once the tree is fitted, it can be queried on axis combinations that were never trained, giving a predicted AUROC for any configuration in the full space. Feature importances from the tree provide a second, independent ranking of axis sensitivity that can be compared directly to the marginal analysis above.

The top predicted configurations are reported as hypotheses, not established results, and feed directly into the recommended model of Ch. 9. No new training is required.

8.4. Recommended configuration

The findings of the marginal analysis and surrogate model are combined into a single recommended axis configuration: each axis is set to whichever value shows a robust improvement, or to the cheapest default where no improvement was found. The recommended configuration is stated explicitly as a Hydra override string and validated in Ch. 9.

9

The Best Model and Its Physics Reach

This chapter does two things. First, it assembles the recommended configuration from the Ch. 8 analysis, trains it at the Pareto-optimal scale from Ch. 9, and compares it against a simple baseline and the external MIParT benchmark. Second, it attempts to translate the AUROC improvement into a physics statement: how much does a better classifier actually matter in units of LHC luminosity?

9.1. The recommended model (Exp. 10A)

The recommended configuration is taken directly from the Ch. 8 surrogate analysis. It is trained at the Pareto-optimal model size from Ch. 9 using five seeds to reduce Monte Carlo uncertainty on the final performance estimate. Two baselines are included for comparison: the simplest possible transformer (T1=`raw`, no biases, smallest preset) and MIParT (Li et al., 2024) evaluated on the same dataset and task.

Model	Description	Runs
Recommended	best config from Ch. 8, Pareto size from Ch. 9	5
Simple baseline	T1=`raw`, no biases, `d32_L3`	3
MIParT reference	Li et al. on same dataset	3
Total		**11**

Table 9.1: Exp. 10A. Both tasks (4t vs. background and 4t vs. ttH) are evaluated for all three models.

Outputs: AUROC and signal efficiency at $1/B = 50$ for all three models on both tasks, with seed-averaged means and error bars. This is the thesis headline number.

9.2. From AUROC to physics sensitivity (Exp. 10B)

Note: this section describes machinery I am still working through and may not appear in the final thesis, depending on whether the results are meaningful and the methodology is sufficiently under control. It is included here as a planning placeholder.

The quantity a physics analysis actually cares about is not AUROC but sensitivity to the four-top-quark production cross-section. The connection is made through a simplified profile likelihood: given the classifier score for each event, a binned discriminant is constructed, signal and background yields are scaled to a target luminosity, and the expected significance is computed under the asymptotic approximation of Cowan et al. (2011). In the absence of systematics this reduces to the familiar form

$$Z \approx \frac{S}{\sqrt{B}}, \tag{9.1}$$

where S and B are the expected signal and background yields at the chosen working point.

The main output is a luminosity-versus-significance curve for the recommended model, the simple baseline, and MIParT. The headline number is $\mathcal{L}_{5\sigma}$, the luminosity at which each classifier first reaches $Z = 5$. A reduction in $\mathcal{L}_{5\sigma}$ directly quantifies how much LHC running time the architectural improvements of this thesis are worth.

All numbers are relative comparisons between classifiers at the same luminosity assumption, using MC yields only with no systematic uncertainties. They should not be read as absolute physics predictions.

10

Conclusion and Outlook

This thesis set out to answer a deceptively simple question: for a transformer-based classifier operating on multi-top quark events at the LHC, which architectural choices matter, and why? Rather than optimising a single model, we built a configurable workbench parameterised along 33 orthogonal axes and used it to run a systematic, controlled ablation programme across roughly one thousand training runs.

The central finding is that the performance of a transformer classifier in this setting is determined by a small number of input-representation choices — principally the tokenisation strategy, the continuous feature set, and whether physics-informed attention biases are active — while the majority of architectural variations imported from the LLM literature, including normalisation policy, attention-internal normalisation, and causal masking, have no statistically distinguishable effect on AUROC. This is itself a practically useful result: it tells a practitioner building a similar classifier exactly where to invest design effort and where defaults are safe. The recommended configuration, derived from the axis findings and validated on both the primary 4t-vs-background task and the harder 4t-vs-$t\bar{t}H$ task, achieves a meaningful improvement in expected discovery significance over the simplest transformer baseline, corresponding to a reduction in the luminosity required to reach $Z = 5$ that is directly quantifiable in units of LHC run time.

Outlook

Several natural extensions follow from this work.

The most immediate is the incorporation of systematic uncertainties into the sensitivity estimates of Chapter 10. The profile likelihood framework used there is already the standard ATLAS machinery; extending it to include b-tagging efficiency, jet energy scale, and luminosity nuisance parameters would close the gap between the benchmark study presented here and a full physics analysis, and would test whether the architectural advantages identified in the axes sweep survive the addition of a realistic uncertainty model.

On the architecture side, the workbench is designed to accept new axes without modifying existing ones. Two additions are of particular interest for future work. First, a graph-based pre-encoder that operates on a dynamically constructed particle graph — rather than the fixed-topology MIA blocks studied in Chapter **??** — would provide a stronger test of whether explicit geometric structure before the transformer encoder is beneficial. Second, the surrogate modelling exercise of Chapter 8 identifies predicted high-performing configurations in the tails of the explored space; training those configurations would close the loop between the statistical analysis and the experimental programme.

Beyond the four-top-quark process, the workbench is process-agnostic by construction. Applying it to three-top production, $t\bar{t}HH$, or other rare multi-object final states at the HL-LHC requires only a new dataset and a redefinition of the class labels, with all architectural findings carrying over directly. The codebase, experiment database, and recommended configuration are publicly available for exactly this purpose.

References

[1] ATLAS Collaboration. “Observation of four-top-quark production in the multilepton final state with the ATLAS detector”. In: *The European Physical Journal C* 83.6 (June 2023). arXiv:2303.15061 [hep-ex], p. 496. ISSN: 1434-6052. DOI: 10.1140/epjc/s10052-023-11573-0. URL: http://arxiv.org/abs/2303.15061.

A

Axes reference table

B

Codebase Struture and Reproducibility

C

Dataset and Feature Definitons